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Microplastic fibers and leachates from cigarette butts: environmental impacts, toxicological concerns, and circular economy-driven solutions

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ABSTRACT

Cigarette butts constitute the most ubiquitous plastic litter worldwide, serving as a persistent source of microplastic fibers and toxic chemical leachates in terrestrial and aquatic ecosystems. Primarily made of cellulose acetate, discarded butts degrade slowly via physicochemical processes, liberating cigarette butt-derived microplastics alongside hazardous additives such as nicotine, heavy metals, and polycyclic aromatic hydrocarbons. This PRISMA-guided narrative review consolidates evidence on the environmental prevalence, transport, and fate of these microplastics, rigorously appraises progress in analytical detection methods, and evaluates their ecotoxicological impacts across food web levels. Studies reveal induction of behavioral changes, elevated mortality, and bioaccumulation in aquatic species, with particles functioning as carriers for persistent organic contaminants and metals. The review also addresses the considerable economic costs of cigarette butt pollution and scrutinizes mitigation options, including cutting-edge removal techniques, extended producer responsibility schemes, and circular-economy pathways such as cellulose acetate reclamation and composite sorbent fabrication. Persistent gaps exist in long-term degradation patterns, realistic exposure limits, and practical remediation efficacy, necessitating focused research to support evidence-based policies and sustainable waste management.

1. Introduction

Global solid waste generation is projected to increase significantly, with annual production expected to rise from 2.01 billion tonnes in 2016 to 3.40 billion tonnes by 2050, according to the World Bank (Oladipupo et al., 2024; Yogaswara et al., 2024). This growth is driven by population growth, urbanization, increased tourism, and changing consumption patterns, all of which contribute to higher volumes of litter entering terrestrial and aquatic ecosystems (Oladipupo et al., 2024; Yogaswara et al., 2024; Yousef et al., 2024). CBs are the most highly generated waste globally. Furthermore, specifically, cigarette butts (CBs) have been identified as a major contributor to the menace of plastic pollution, leading to environmental contamination, with around 4.5 trillion CBs being discarded annually (Valiente et al., 2020; Slaughter et al., 2011). In 2016, global cigarette production reached 5.7 trillion, predominantly composed of cellulose acetate (CA), a non-biodegradable synthetic polymer that releases microplastics (CBMPs) and toxic leachates into ecosystems (Yousef et al., 2024; Díaz-Mendoza et al., 2024; Venugopal et al., 2021). CBs are not only a source of microplastic fibers but also release over 7000 toxic substances, including nicotine, a neurotoxic chemical that disrupts neurotransmitter release in the autonomic and central nervous systems (Valiente et al., 2020; Slaughter et al., 2011; Díaz-Mendoza et al., 2024; Venugopal et al., 2021). Prolonged activation of nicotinic acetylcholinergic receptors (nAChRs) may lead to receptor desensitisation, impairing brain function (Wright et al., 2015). Additionally, at least 250 harmful compounds, including 69 carcinogens, are released as CBs degrade, contaminating soil and water systems (Zeng et al., 2023), (Ajibade et al., 2023). For example, discarded cigarette butts can contribute to water contamination, with smoked filters showing higher toxicity than unsmoked filters owing to the accumulation of harmful substances during smoking (Green et al., 2014), (Slaughter et al., 2011; Belzagui et al., 2021).

The ecological and health impacts of CB pollution are profound.

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Abbreviations

MPs	microplastics
CB	cigarette butt
CBs	cigarette butts
CBMPs	cigarette butt-derived microplastics
PAHs	polycyclic aromatic hydrocarbons
SFs	smoked filters
MFs	microfibers.

Considering the environmental and health hazards associated with CBs, it is imperative to devise a sustainable and effective solution to address CB pollution. This will help protect ecosystems and human well-being.

Despite the expanding body of research on cigarette butt pollution, existing reviews remain narrowly focused on chemical toxicity profiles or direct human health risks associated with leachates. Notably absent are comprehensive syntheses that integrate the growing evidence base on cigarette butt–derived microplastics (CBMPs) across their full environmental lifecycle from occurrence and persistence in diverse matrices (water, soil, sediments), to analytical characterisation challenges, ecotoxicological effects on aquatic and terrestrial biota, and scalable remediation or valorisation strategies. This fragmented literature landscape leaves critical knowledge gaps regarding CBMPs as a distinct microplastic subtype, particularly their transport dynamics, long-term environmental fate, and potential for integration into circular economy strategies. Existing literature also underemphasizes systems-level management approaches capable of transforming CBMP-laden waste streams into recoverable resources.

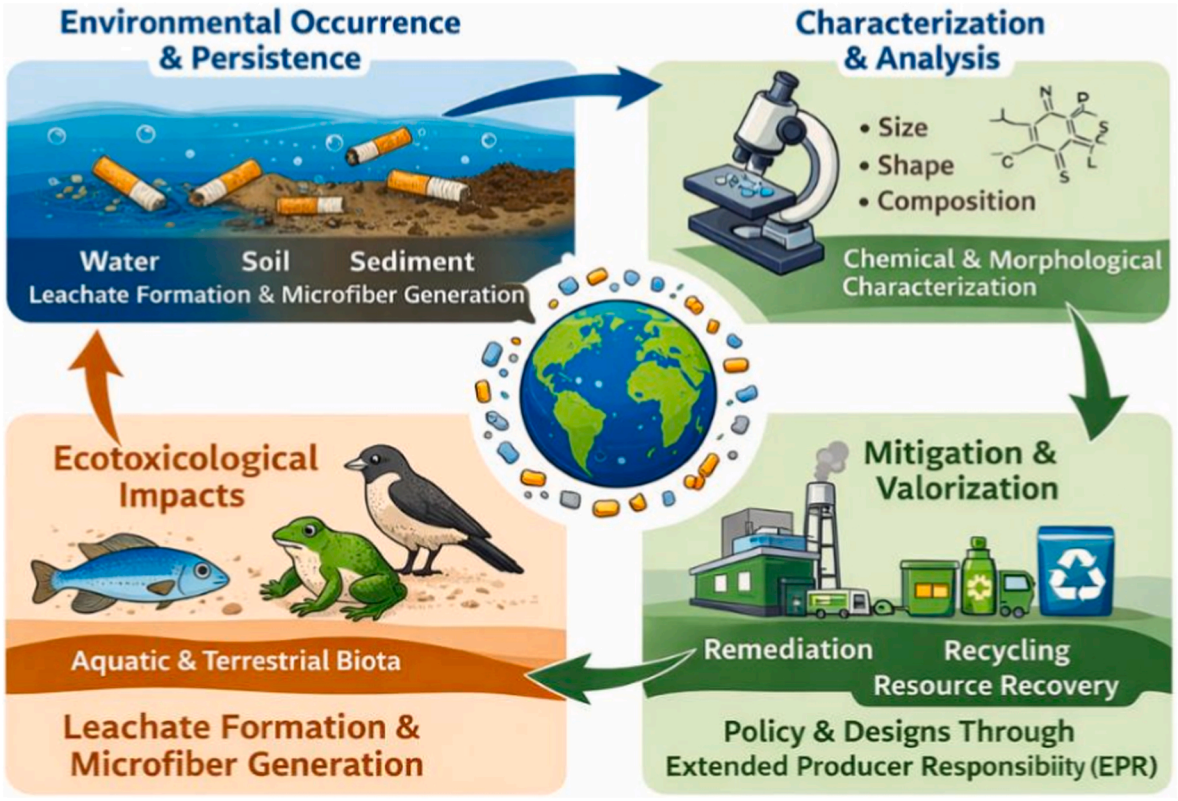


Fig. 1. An integrated systems perspective on cigarette butt–derived microplastics: From environmental fate and toxicity to policy and circular economy solutions.

Freshwater and marine species exposed to CB leachates exhibit altered behavior, reduced survival rates, and bioaccumulation of toxins, which can cascade through food webs (Hossaini Motlagh et al., 2024; Booth et al., 2015). Besides affecting wildlife, CB contamination poses indirect risks to human health by consuming contaminated crops or animals (Yang et al., 2024; Green et al., 2020). For instance, birds that use CBs as nesting material expose their young to dangerous chemicals, which may cause long-term genetic damage and other health risks (Kaplan, 2012). Furthermore, male birds that stayed longer in the nest, caring for the chicks, were found to have suffered genotoxic damage, indicating that prolonged exposure to these compounds poses significant risks (Suárez-Rodríguez et al., 2017). It is known that aquatic species, such as fish, turtles, whales, and birds, often ingest CBs, which can disrupt their digestive systems and lead to poisoning or starvation (Oceancare; Novotny et al., 2011). Furthermore, recent studies indicate that CB-derived nanoplastics are absorbed and dispersed throughout ecosystems, presenting further ecological threats (Yang et al., 2024; Chevalier et al., 2018). Urban and coastal areas, especially beaches, exhibit high levels of CB pollution due to heavy human activity and proximity to cities (Venugopal et al., 2021; Novotny et al., 2009).

This PRISMA-guided narrative review addresses these gaps by systematically synthesizing evidence across four interconnected domains: (i) environmental occurrence, distribution, and persistence; (ii) physicochemical and morphological characterisation; (iii) ecotoxicological impacts across trophic levels; and (iv) mitigation technologies coupled with circular economy valorisation strategies and policy and design through extended producer responsibility (EPR).

By bridging these siloed domains, this review provides actionable insights for policymakers, waste managers, and researchers addressing the global CBMP pollution crisis. Fig. 1 summarizes the scope of this study by linking environmental occurrence, characterisation, ecotoxicological impacts, and mitigation and policy pathways for cigarette butt–derived microplastics.

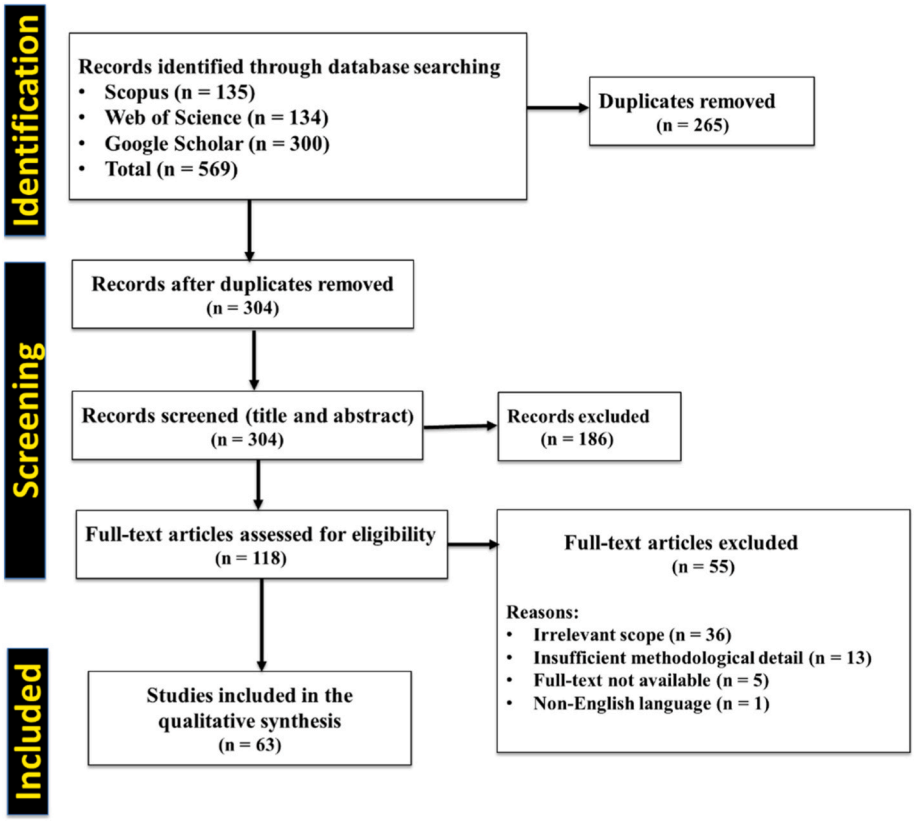


Fig. 2. PRISMA-guided flow diagram of the systematic literature screening and selection process for the cigarette butt–derived microplastics study.

2. Methodology

This review adopts a PRISMA-guided narrative synthesis approach to ensure transparency, reproducibility, and broad coverage of the literature, while retaining flexibility appropriate for a multidisciplinary environmental review. The methodology was designed to systematically identify, screen, and synthesise peer-reviewed studies on cigarette butt–derived microplastics (CBMPs), associated leachates, ecotoxicological impacts, analytical characterisation methods, and circular-economy-based mitigation strategies.

2.1. Literature search strategy

A comprehensive literature search was conducted across multiple scientific databases to minimise selection bias and maximise coverage. The primary databases consulted were Scopus, Web of Science Core Collection, and Google Scholar, covering the period January 2015 to February 05, 2026. This time window was selected to capture both early foundational studies on cigarette butt pollution. Search strings were adapted to the syntax and indexing structure of each database while preserving the same core conceptual framework (pollution source, plastic characteristics, and environmental impact). Minor variations in keywords were necessary to optimise retrieval efficiency and relevance across platforms, particularly for Google Scholar, which has limitations on complex Boolean queries. Google Scholar returned approximately 1610 search results. Consistent with established practice for systematic and scoping environmental reviews, only the first 300 relevance-ranked records were screened, as results beyond this threshold showed diminishing relevance and high redundancy. Google Scholar's ranking algorithm prioritizes citation relevance, and screening a defined subset improves feasibility while maintaining coverage of the most influential literature. Accordingly, the PRISMA flow diagram reflects the 300 records that were actually screened rather than the total raw search output (See Fig. 2), while Table 1 Represents the summary of the database search strategy and results adopted in this study.

2.2. Study screening and eligibility criteria

All references identified through database searching were imported into EndNote 21 (Clarivate Analytics) for reference management. Duplicate records were identified and removed using EndNote's duplicate detection function. Titles and abstracts were screened manually against predefined inclusion and exclusion criteria, followed by full-text eligibility assessment. Full texts were evaluated systematically, and studies not meeting the eligibility criteria were excluded with documented reasons. Study selection was conducted in accordance with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines. A two-stage screening process was employed.

1. Title and abstract screening, followed by
2. Full-text eligibility assessment

Studies were included if they met the following criteria.

- Peer-reviewed journal articles written in English
- Focused on cigarette butts, cigarette filters, or cigarette butt-derived microplastics
- Reported data on environmental occurrence, fate, analytical characterisation, leachate chemistry, ecotoxicological effects, or mitigation/valorisation strategies
- Experimental, field-based, modelling, or review studies relevant to environmental systems

Studies were excluded if they.

Table 1
Detailed summary of the database search strategy, including Boolean search strings, time range, and number of records retrieved from Scopus, Web of Science, and Google Scholar.

Database	Boolean search strings	Time range employed for the search	Retrieved records
Scopus	("cigarette butt*" OR "cigarette filter*" OR "tobacco filter*" OR "tobacco waste') AND (microplastic* OR microfiber* OR nanoplastic* OR "cellulose acetate') AND (leachate* OR toxicity OR ecotoxicology OR "environmental impact*" OR bioaccumulation OR "aquatic organism*" OR "circular economy' OR remediation OR recycling OR "extended producer responsibility")	2015–Feb 2026	135
Web of Science	TS= ("cigarette butt' OR "cigarette filter' OR "tobacco filter' OR "tobacco waste') AND (microplastic OR microfiber OR nanoplastic OR "cellulose acetate') AND (leachate OR toxicity OR ecotoxicology OR "environmental impact' OR bioaccumulation OR "aquatic organism' OR "circular economy")	2015–Feb 2026	134
Google Scholar	("cigarette butt' OR "cigarette filter') AND (microplastic OR microfiber OR nanoplastic OR "cellulose acetate') AND (leachate OR toxicity OR ecotoxicology OR "environmental impact' OR "circular economy') (Note: Top 300 results screened for relevance)	2015–Feb 2026	300
Total	—	—	569

- Focused exclusively on human smoking behaviour or clinical health outcomes
- Addressed tobacco smoke chemistry without environmental relevance
- Lacked primary data or substantive analysis (e.g., editorials, opinion pieces)

Disagreements during screening were resolved through repeated discussion and reassessment of the eligibility criteria until consensus was achieved.

2.3. PRISMA-style study selection overview

The literature identification, screening, and inclusion process is summarised using a PRISMA-style flow diagram (Fig. 2). While this review is not a formal systematic review or meta-analysis, the PRISMA framework was applied to enhance transparency and reproducibility of the evidence synthesis process.

2.4. Data extraction and thematic synthesis

From each included study, the following information was extracted.

- Publication year and geographic focus
- Environmental matrix (water, sediment, soil, biota)
- Type of cigarette butt material (smoked, unsmoked, biodegradable)
- Microplastic characteristics (size, morphology, polymer type)
- Analytical techniques employed (e.g., SEM, FTIR, ICP-MS)
- Ecotoxicological endpoints and test organisms
- Mitigation, recycling, or circular economy applications

Extracted data were synthesised thematically across four domains.

(i) environmental occurrence and fate,
(ii) physicochemical and analytical characterisation,
(iii) ecotoxicological impacts, and
(iv) mitigation and circular economy-driven solutions.

2.5. Quality appraisal and bias considerations

Given the heterogeneous nature of the included studies, a formal exclusion-based risk-of-bias scoring was not applied. Instead, a qualitative quality appraisal was undertaken to inform the interpretation of findings.

i. For ecotoxicological studies, attention was given to:
   - Use of environmentally relevant exposure concentrations
   - Clear reporting of control conditions and replication
   - Appropriateness of test organisms and endpoints
ii. For analytical and characterisation studies, the appraisal focused on:
   - Method validation and detection limits
   - Use of complementary techniques (e.g., SEM with FTIR or EDS)
   - Transparency in sample preparation and contamination control
iii. For mitigation and circular economy–driven solution studies, attention was given to:
   - Clarity in describing the intervention or recovery approach (e.g., EPR, material recovery)
   - Assessment of environmental and economic feasibility or scalability
   - Evidence of demonstration or validation under realistic conditions

Studies with methodological limitations were not excluded but were explicitly contextualised in the discussion to avoid over-interpretation. This approach is consistent with best practice for narrative environmental reviews and allows integration of emerging evidence in a rapidly evolving research field.

2.6. Summary of included studies

An overview of 11 representative studies is presented in Table 2, summarising key attributes, including study focus, environmental matrix, analytical or experimental methods, and principal findings, to highlight methodological diversity and provide readers with insight into the scope of included research. The full set of studies included in this PRISMA-guided review is n = 63 (see Fig. 2 and Table 1).

2.7. Operational definitions for microplastics (MPs) and nanoplastics (NPs)

Cigarette butt microplastics (CBMPs) are operationally defined as cellulose acetate microfibers originating from degraded cigarette filters and confirmed using polymer-specific analytical methods to distinguish them from natural or other synthetic fibers. Details of operational definitions adopted in this study are presented in Table 3.

2.8. Methodological limitations and scope considerations

This review was designed as a scoping, PRISMA-informed narrative synthesis to map and integrate the rapidly expanding literature on cigarette butt–derived microplastics (CBMPs), their environmental occurrence, physicochemical characteristics, ecotoxicological effects, and emerging mitigation and circular economy–oriented responses. As such, the primary objective was to provide a comprehensive overview of evidence domains and research trends, rather than to quantitatively evaluate effect sizes or establish causal relationships. Consistent with

Table 2
Summary of key representative studies included in this PRISMA-guided review on cigarette butt–derived microplastics and leachates.

Study type	Primary focus	Environmental matrix/ Applications	Analytical/ experimental methods	Key findings	Relevance to this review	Ref.
Laboratory ecotoxicology	Microfiber release from smoked cigarette filters	Freshwater (leachate)	SEM, microfiber counting, acute toxicity assays (*Daphnia magna*)	Smoked filters release ~100 microfibers/day; high toxicity observed at low SF/L concentrations	Demonstrates CBs as a significant source of toxic microplastic fibers	Belzagui et al. (2021)
Field survey + lab analysis	CB-derived MPs and heavy metals in urban environments	Urban surfaces, runoff water	SEM–EDS, ICP-MS	CBs contribute significant microfiber and heavy metal loads in cities	Links CBMP occurrence with urban hydrology and metal contamination	Yang et al. (2024)
Laboratory ecotoxicology	Toxicity of MPs from smoked vs unsmoked filters	Freshwater & soil organisms	Milling of filters, SEM, particle size analysis, survival assays	Smoked filter MPs more toxic than unsmoked; metals detected on MPs	Highlights differential toxicity and contaminant vector potential	Dolar et al. (2025)
Chemical characterisation	PAHs and metal leaching from CBs	Water (leachate)	GC-MS (PAHs), ICP-MS (metals)	Multiple PAHs and metals leached above regulatory thresholds	Confirms CB leachate as a chemical hazard beyond plastics alone	Mušič et al. (2023)
Field + mesocosm study	Fate and toxicity of CBs in coastal systems	Coastal water and sediment	Settling experiments, copepod toxicity tests	Adverse effects observed at ≤0.1 CB/L in water	Demonstrates the real-world ecological relevance of CB pollution	Freire Lima et al. (2021)
Developmental ecotoxicology	Teratogenic effects of CB leachates	Freshwater (embryo assays)	FETAX (*Xenopus laevis* embryos)	Smoked CB leachate caused deformities at <1 CB/L	Supports chronic toxicity concerns of CB leachates	Parker et al. (2017)
Analytical methods study	Morphology of CA microfibers	Cigarette filters	SEM, ImageJ fiber measurement	Significant variability in fiber size across brands	Supports challenges in CBMP characterisation	Soltani et al. (2025)
Degradation study	Structural changes in biodegradable vs CA filters	Soil/simulated environments	SEM, controlled degradation experiments	Biodegradable filters altered microbial communities	Challenges assumptions about "eco-friendly" filters	Parida et al. (2025)
Materials synthesis and recovery	Upcycling of CB into carbon materials for remediation	Materials in wastewater treatment	SEM, FITR	Materials synthesis and waste management	Circular economy approach	Adeleye et al. (2025)
Applied study on materials recovery	CB recovery and valorisation	Materials in wastewater treatment	Material recovery, composite fabrication	CA recovered for sorbents/ composites	Demonstrates the circular economy potential of CB waste	(De Cesaris et al., 2024; Janković et al., 2023)
Policy and designs	Mitigation by Extended Producer Responsibility (EPR)	Waste control	Design, policy, and enforcement	Accountability and policy enforcement	Mitigation by policy enforcement	Croes et al. (2023)

Table 3
Definitions table (box 1).

Term	Size/Morphology Criteria	Measurement/Identification Method	Notes	Ref.
Microplastic (MP)	1 μm –< 5 mm	Optical microscopy, FTIR, Raman spectroscopy	Includes fibers, fragments, films, and beads	(Belzagui et al., 2021; El Hadri et al., 2021)
Nanoplastic (NP)	<1 μm	Electron microscopy (SEM/TEM), Raman, AFM	The current analytical resolution limits detection	(Mušič et al., 2023; El Hadri et al., 2021)
Fiber (including CBMPs)	Aspect ratio ≥3:1, length ≥1 mm	Stereomicroscopy, SEM, FTIR, Raman	Specifically includes cellulose acetate fibers from cigarette filters	(Belzagui et al., 2021; Acarer Arat, 2024)
Fragment	Irregular shape, length <5 mm, aspect ratio <3:1	Optical microscopy, FTIR	Distinct from fibrous CBMPs	(Belzagui et al., 2021; Mušič et al., 2023)
Cigarette Butt Microplastic (CBMP)	Fiber derived from cigarette filter; predominantly cellulose acetate; length 1–5 mm	Optical microscopy and FTIR confirmation of cellulose acetate, Pyrolysis–GC/MS (for polymer-specific confirmation), μ-FTIR mapping (for complex environmental matrices)	Distinguished from natural fibers and other anthropogenic fibers	(Soltani et al., 2025; Shen et al., 2021)

Standard microplastic size is plastic particles <5 mm (often ≥1 μm lower bound), with nanoplastics <1 μm (Soltani et al., 2025; Shen et al., 2021). Fiber classification uses aspect ratio ≥3:1 (length:diameter) (Mušič et al., 2023; Shen et al., 2021). CBMPs are confirmed as cellulose acetate microfibers from degraded filters, fitting fiber criteria and <5 mm (Shen et al., 2021; Sharma et al., 2025).

the remit of scoping reviews, no formal risk-of-bias assessment or quantitative quality scoring of individual studies was conducted. The included literature spans highly heterogeneous study designs, including laboratory-based ecotoxicological assays, field surveys, analytical method development studies, and conceptual or policy-oriented analyses. Applying a single standardized bias appraisal tool across these diverse study types was, therefore, considered methodologically inappropriate and potentially misleading. Nevertheless, study limitations, methodological constraints, and uncertainties reported by the original authors have been critically considered in the interpretation and synthesis of findings. The review is further limited to peer-reviewed literature published in English and indexed in major bibliographic databases, which may have resulted in the exclusion of relevant studies reported in other languages or in the grey literature. In addition, while this review integrates evidence across multiple environmental matrices and biological endpoints, variability in analytical methods, exposure conditions, and reporting standards restricts direct comparability across studies.

Finally, although mitigation and circular economy–based strategies for cigarette butt waste are discussed, the review does not constitute a life-cycle assessment or a quantitative environmental performance evaluation of these approaches. Consequently, proposed solutions are

Table 4
Classification of CB based on hazardous properties (HP).

Hazardous Properties (HP) Classification	Highlights of the adverse impacts of cigarette butts according to each HP	Ref.
HP 4 (Irritant)	The presence of NH_3 and other aldehydes, which have been reportedly found in cigarette smoke which may have an impact on the eyes and respiratory system.	(Watson et al., 2016; Seeman, 2007)
HP 5 (Harmful)	Nicotine and its metabolites, carbon monoxide, thiocyanate, and some aromatic amines are all capable of exerting cardiovascular damage.	Leone (2015)
HP 6 (Toxic)	A study has shown that exposure to cigarette smoke via the respiratory system harmed the uterus and ovary in female rats under induced diabetes. Therefore, the authors concluded that the potential adverse effects of diabetes can include impaired reproductive health in women	Kim et al. (2019)
HP 7 (Carcinogenic)	Out of over 5000 compounds identified in cigarette smoke, over 73 compounds have been tagged carcinogenic according to either laboratory animals or humans by the International Agency for Research on Cancer	(Rodgman et al., 2008; Hecht, 2011)
HP 10 (Toxic for reproduction)	Cigarette smoking impedes the ability to become pregnant and carry a pregnancy to term.	(Rebischung et al., 2018b; Rzymski et al., 2015)
HP 11 (Mutagenic)	The multiple carcinogens reported in cigarette smoke have been identified as one of the major factors responsible for the complex mutations observed in critical cancer genes.	Hecht et al. (2022)
HP 13 (Sensitizing)	Maternal smoking, particularly in utero, has been linked to an increased risk for the later development of childhood atopy and asthma. Furthermore, it is important to state that smoking aggravates the inflammatory burden of the lower respiratory tract.	Floreani et al. (1999)
HP 14 (Ecotoxic)	Both leachates and microplastic fibres generated from CBs are detrimental	Venugopal et al. (2021)

framed as risk-mitigation and resource-recovery pathways rather than definitive pollution-prevention outcomes. These limitations highlight the need for future research employing harmonised methodologies, long-term field assessments, and integrated environmental and techno-economic analyses to strengthen evidence-based decision-making.

3. Global burden and source of CBMPs

Improperly disposed of CBs release toxic compounds, polluting terrestrial and aquatic ecosystems. Studies have shown that these discarded filters leach hazardous chemicals, presenting substantial threats to ecological integrity and human health.

3.1. Classification of CB based on hazardous properties (HP)

The European Waste Framework Directive regards CBs as hazardous waste. More specifically, they belong to the HP 6 (toxicity) and HP 14 (ecotoxicity) groups (Rebischung et al., 2018a). Their nicotine concentration and the harmful chemicals they release into the environment define this classification mostly (Zeng et al., 2023; Rebischung et al., 2018a). The full classification is presented in Table 4, along with brief highlights. This classification raises awareness of more stringent laws and environmentally suitable disposal practices, and mitigation strategies to safeguard human health and the environment.

3.2. Biodegradable CBs: environmental burdens and misconceptions

Biodegradable cigarette filters have recently been suggested as alternatives to traditional cellulose acetate (CA) filters to address environmental concerns. Historically, cigarette filters were first introduced between 1860 and 1920 to prevent tobacco particles from entering the smoker's mouth (George et al., 2022). Cellulose acetate filters have been commercially available since 1927, originating with filter-tipped cigarettes in Switzerland (Pauly et al., 2009). Following the early 1950s 'lung cancer scare', cigarette manufacturers heavily invested in filter-tipped cigarette design and marketing (Zipser, 1954). Consumers widely perceived these 'filters' as reducing harmful smoke constituents, with Congressman John Blatnik noting in 1957 that the term "filter' carried substantial preconceived efficacy (Harris, 2011). This public trust created a critical window for tobacco engineers to develop genuinely effective filtration, though industry scientists soon recognized the inherent challenges, coining the term "filter problem' by 1954 (Klamkin, 1957). Despite these efforts, modern cellulose acetate filters, now ubiquitous, proved largely ineffective at reducing health risks and posed persistent environmental issues due to their slow degradation in natural settings (Vert et al., 2012). According to the European Commission's standards (EN 13432), true biodegradability requires compostable materials to convert at least 90% into CO_2 under microbial action within six months, a threshold unmet by conventional cigarette filters (Innocenti and Biodegradability, 2018). Therefore, this characteristic feature is validated with the aid of a series of standardized laboratory test procedures (Koster, 2025). This has, however, generated heated debate about the biodegradability of CBs, which is due to the conducted assessment under controlled laboratory conditions and differs from the complex real-world environmental representations (Vert et al., 2012; Koster, 2025). The actual degradation of CBs occurs through a combination of biodegradation and photodegradation processes (Koster, 2025).

It is important to emphasize that the ecotoxicological impact of biodegradable filters is complex and of concern. For instance, a recent study by Koroleva and colleagues found that although both biodegradable and non-biodegradable CBs contain different heavy metals, biodegradable CBs more significantly altered bacterial community composition. This was attributed to their higher levels of leached hazardous metals (Koroleva et al., 2021). The authors further employed automated ribosomal intergenic spacer analysis (ARISA) for the evaluation of bacterial diversity before and during CB leachate exposure. The reported findings indicated that biodegradable CBs significantly altered the architecture of bacterial communities more than non-biodegradable ones. These findings further highlight the detrimental effects of CB pollution on microbial communities, which are critical for maintaining soil health. Consequently, the inference drawn from their study confirmed that biodegradable CBs may not be as environmentally friendly as previously thought (Koroleva et al., 2021).

This misconception may also lead to increased pollution due to a false sense of environmental safety. Smokers are likely to believe these filters are harmless, making them more likely to be disposed of inappropriately. Therefore, biodegradable cigarette filters may not be the answer they initially appear to be. The persistence of hazardous chemicals, the potential for increased CB waste, the formation of CBMPs, and the lack of clarity surrounding the ecotoxicological impacts suggest that these alternatives may result in new environmental issues. For example, a study on bioplastic MPs (poly (hydroxybutyrate) (PHB) and poly (lactic acid) (PLA)) showed ecotoxicological effects on *Daphnia magna* similar to those of conventional MPs (HDPE), in terms of behavioural

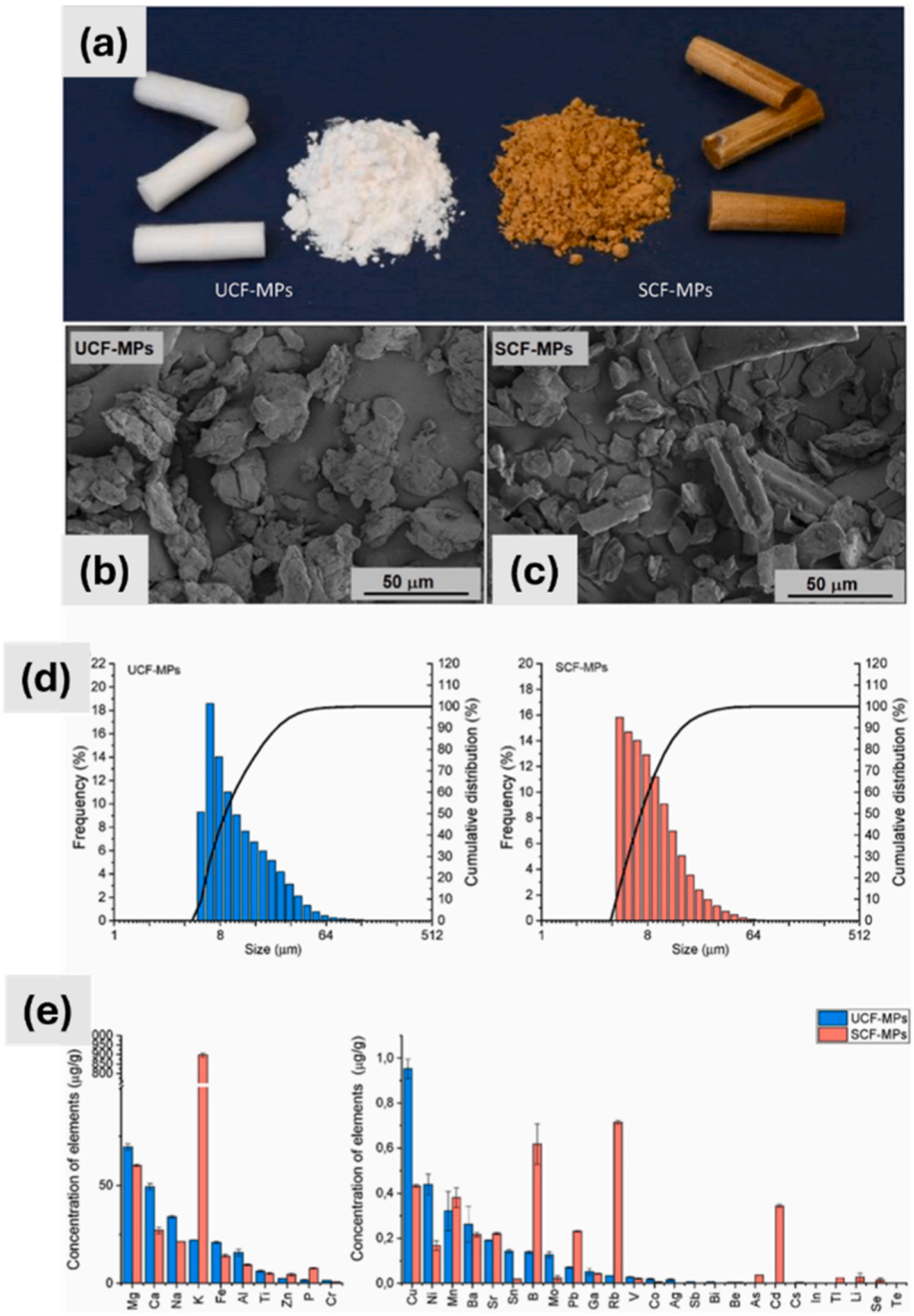


Fig. 3. (A). Characteristics of microplastics milled from unsmoked (UCF-MPs) and smoked cigarette filters (SCF-MPs). (b) Scanning electron microscope of UCF-MPs & SCF-MPs. (d). Particle size distribution of UCF-MPs & SCF-MPs (e). Metals and other selected elements detected in concentration ranges >1.5 µg/g (left), and <1.5 µg/g (right) (Dolar et al., 2025) with permission. Copyright 2025 Elsevier.

and feeding patterns. This emphasizes the necessity of conducting in-depth studies on their long-term ecological impacts. Furthermore, in another study, a six-month experiment under various environmental conditions investigated the degradation of cellulose and CA filters. The findings showed that 33.5% and 83.1% were the final masses of the cellulose and CA filters, respectively. The remaining masses reported for cellulose and CA filters clearly revealed that the rate of degradation observed for cellulose is faster than that of CA. In addition, the authors further comparatively investigated the mass loss of smoked vs unsmoked filters; results showed that the higher mass losses were observed for CA filters, which was attributed to leaching, while smoked filters were more vulnerable to the highest mass losses. The authors then drew an inference that the CA filters take approximately 7.5 – 14 years to degrade, while smoked cellulose filters take about 2.3 – 13 years to degrade (Joly et al., 2018). Smoked cigarette filters pose greater environmental and ecotoxicological hazards than non-smoked (unused) filters due to absorbed toxins from tobacco smoke. Studies show leachates from smoked filters cause higher mortality and sublethal effects in aquatic organisms compared to unused ones (Slaughter et al., 2011). Furthermore, laboratory bioassays reveal a clear toxicity gradient: smoked butts > smoked filters > unsmoked filters. For marine topsmelt (*Atherinops affinis*), LC50 values (concentration killing 50% of test subjects) are 1.1 butts/L for smoked butts, 1.8 butts/L for smoked filters (no tobacco), and 5.1 butts/L for unsmoked filters. Freshwater fathead minnows (*Pimephales promelas*) show similar patterns: 0.97 butts/L (smoked butts), 4.3 butts/L (smoked filters), and 13.5 butts/L (unsmoked filters) (Green et al., 2020). In addition, in medaka fish (*Oryzias latipes*) embryos, leachates ranked by toxicity as: smoked tobacco > unsmoked

tobacco > smoked filters > unsmoked filters, with effects like altered heart rate, suppressed development, and increased mortality starting at ≤0.2 pieces/L; unsmoked filters showed no effects up to 20 pieces/L (Lee et al., 2015). Therefore, the environmental impact arises from discarded cigarette waste, which releases nicotine, heavy metals, and combustion-derived compounds into surrounding water and soil environments via leaching processes (Green et al., 2020; Shen et al., 2021; Lee et al., 2015).

4. Environmental fate and impacts of CB-MPs

4.1. Potential generation of microplastic fibers from CBs

Cigarette filters are designed to capture harmful compounds and particles found in tobacco smoke and are commonly made from cellulose acetate (CA). The continuous release of microplastic fibers from CBs into the environment can be triggered by physical, chemical, and biological breakdown processes (Hon, 1977; Ach, 1993). A recent study examined how microfibers (MFs) generated from cigarette-smoked filters (SFs) contribute to MPs pollution. According to their study, synthetic fibers containing 15,000 strands release around 100 microfibers (MFs) smaller than 0.2 mm daily. An estimated 0.3 million tons of microfibers could be added to aquatic environments each year as a result of this phenomenon. Additionally, the authors established that MFs in SF leachate demonstrated high toxicity to *Daphnia magna*, with EC50 and EC100 values of 0.017 SF/L and 0.620 SF/L, respectively. Their study clarified the high frequency of synthetic polymers found in deep-sea sediments by showing that microfibers made of synthetic materials constitute a significant source of total MPs (Belzagui et al., 2021).

Another study conducted in Dalian, China, examined 35 distinct cigarette brands and 10,591 CBs. The analysis estimated that Chinese CBs contribute around 360,000 tonnes of plastic waste yearly. Recently, the discharge and contamination levels of MFs were examined across four Chinese cities. CB contamination varied, with Guangzhou having 0.03 ± 0.02 CBs/m^2, Meizhou 0.07 ± 0.02 CBs/m^2, Baoding 0.06 ± 0.02 CBs/m^2, and Dalian 0.10 ± 0.03 CBs/m^2. ICP-MS detected 629.7 μg/L of 14 heavy metals, while SEM and EDS investigations showed the presence of MPs and heavy metal particles in CBs, with an estimated 1.9 ± 0.9 g/km^2/day leaking into the environment (Yang et al., 2024). The vector potential of CB-derived MPs has also been reported recently. In one study, CA fibers from cigarette filters were also examined as sources of trace metals in coastal waterways. The fibers with release rates ranging from 40% to 90% were observed by the authors. These results highlight the environmental role of cellulosic MFs in transporting metals in aquatic systems (Turner et al., 2024).

Generally, MPs can be produced by both smoked and unsmoked cigarette filters (Acarer Arat, 2024). Researchers milled MPs from smoked and unsmoked cigarette filters to mimic environmental generation, which was evaluated to quantify their toxic effects on a few chosen aquatic and terrestrial species. After 48 h of exposure, *D. magna* (100 mg L^{-1} of SCF-MPs) and *B. calyciflorus* (10 mg L^{-1}) showed significantly reduced survival, with SCF-MPs exhibiting more severe effects than UCF-MPs. Heavy metal levels were also assessed in both UCF-MPs and SCF-MPs (Fig. 3). Their research indicates the need for efficient environmental control to reduce CB pollution from both smoked and unsmoked cigarettes (Dolar et al., 2025).

4.2. Associated chemical contaminants: heavy metals and polycyclic aromatic hydrocarbons (PAHs)

Beyond their role as microplastic sources, cigarette butts (CBs) act as vectors for chemical contaminants, including heavy metals (e.g., cadmium, lead, arsenic) and polycyclic aromatic hydrocarbons (PAHs) (Yousef et al., 2024; Dobaradaran et al., 2020). These compounds originate from both the tobacco and additives in filters, and can leach into soil, sediments, and aquatic systems upon disposal. Leachate studies indicate that heavy metals and PAHs are readily mobilized under environmental conditions, potentially impacting microbial communities and higher trophic levels (Yousef et al., 2024; Yang et al., 2024). While these contaminants are distinct from the microplastic fraction, their co-occurrence underscores the multifaceted environmental hazard posed by CBs, necessitating integrated management strategies that address both physical and chemical pollution pathways (Acarer Arat, 2024). In a recent study, heavy metal analysis revealed that the most contaminated brand contained 17 metals, while the least contaminated brand contained 15, with concentrations ranging from 0.08 mg/L to 297 mg/L. Nine PAHs were detected in both brands, with concentrations ranging from 0.40 ng/L to 92.12 ng/L (Yang et al., 2023a). The ageing effects on CA obtained from CBs were investigated in a separate study. The authors confirmed that CA fragmentation into MPs is attributed to the ageing effect. Furthermore, CBs also pollute soil and marine ecosystems by releasing toxic substances such as nicotine, arsenic, and heavy metals, as reported by the authors (Mušič et al., 2023). Recently, the discharge and contamination levels of MFs were examined across four Chinese cities. CB contamination varied, with Guangzhou having 0.03 ± 0.02 CBs/m^2, Meizhou 0.07 ± 0.02 CBs/m^2, Baoding 0.06 ± 0.02 CBs/m^2, and Dalian 0.10 ± 0.03 CBs/m^2. ICP-MS detected 629.7 μg/L of 14 heavy metals, while SEM and EDS investigations showed the presence of MPs and heavy metal particles in CBs, with an estimated 1.9 ± 0.9 g/km^2/day leaking into the environment (Yang et al., 2024). The vector potential of CB-derived MPs has also been reported recently. In one study, CA fibers from cigarette filters were also examined as sources of trace metals in coastal waterways. The fibers with release rates ranging from 40% to 90% were observed by the authors (Turner et al., 2024). These results highlight the environmental role of cellulosic MFs in transporting metals in aquatic systems (Acarer, 2023).

4.2.1. Evidence-based human exposure routes to CBMPs

Humans are exposed to microfibers and associated chemicals primarily through ingestion and inhalation, with secondary pathways including dermal contact (Yang et al., 2023b; Savva et al., 2023). Ingestion occurs via contaminated seafood such as fish and bivalves, where microfibers from degraded filters accumulate through aquatic food webs (Novotny et al., 2011; Dolar et al., 2025). Drinking water further contributes to oral exposure, as microfibers persist in urban runoff and wastewater (Hecht et al., 2022; Yang et al., 2023b). Inhalation represents a major route in indoor and outdoor settings, with airborne cellulose acetate fibers from littered butts likely penetrating the lungs (Yang et al., 2023b; Talhout et al., 2024). Particles as small as a few micrometres can translocate into the bloodstream and tissues (Yang et al., 2023b; Everaert et al., 2023). Dermal exposure, though less quantified, occurs through direct contact with littered CBs in public spaces, potentially leading to incidental ingestion (Araújo et al., 2022; Rochman et al., 2013). While systemic detection of microplastics in human placenta, lungs, and blood confirms ubiquitous exposure, no studies have specifically quantified CBMP intake, despite their prevalence as the most littered plastic item globally, plausibly amplifying total microplastic burden across these pathways.

4.2.2. Cigarette butt specific additives and leachate exposure

Leachates from CBs can release nicotine, heavy metals such as lead, cadmium, and chromium, polycyclic aromatic hydrocarbons (PAHs), and aromatic amines, all of which are recognized toxicants with documented human health effects at relevant doses (Yousef et al., 2024; Booth et al., 2015; Dobaradaran et al., 2020). The cellulose acetate fibers themselves are prevalent in the environment and can act as vectors for sorbed chemical contaminants, which may be mobilized through weathering (Novotny et al., 2011; El Hadri et al., 2021). While direct measurements of CB-derived microplastics in human tissues remain limited, the combination of physical particles and associated chemical load underscores the plausibility of cigarette butts as a contributor to human microplastic exposure beyond the background environmental

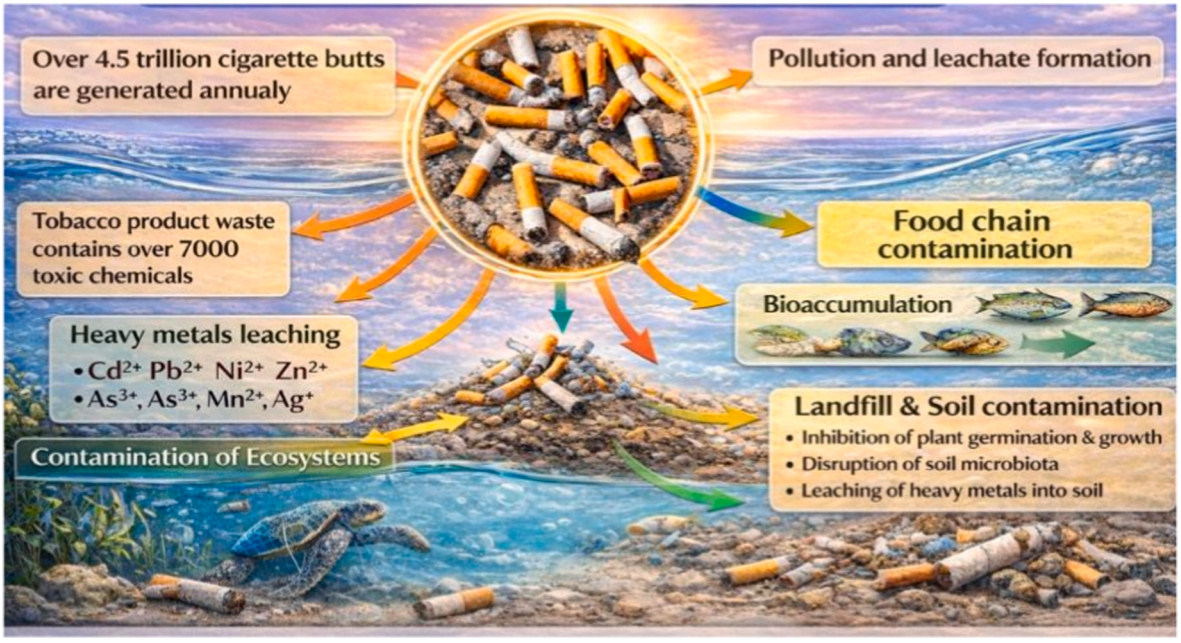


Fig. 4. Fates of cigarette butts into the environment.

burden.

4.2.3. Evidence quality and uncertainty

Despite established pathways of exposure, causal links between CBMP exposure and human health outcomes remain uncertain due to sparse epidemiological data. Epidemiological evidence is currently limited, with systematic reviews suggesting possible impacts on digestive, respiratory, and reproductive health, but the overall quality of human data is low (Zeng et al., 2023; Sharma et al., 2025; Choudhary et al., 2024). Direct evidence quantifying human intake of CB-derived microplastics or leachate chemicals is currently scarce, representing a significant knowledge gap. Regulatory and methodological challenges, including variability in analytical methods, detection limits, and the lack of standardized exposure metrics, further complicate comprehensive health risk assessment.

4.3. The hydrological implications of CBMPs

Hydrological studies indicate that human activity and land use patterns affect the generation and distribution of CBMPs (Suárez-Rodríguez et al., 2017; El Hadri et al., 2021). Discarded CBs release MPs through multiple environmental processes (Ajibade et al., 2023; Belzagui et al., 2021; Green et al., 2020). Due to their size, intact cigarette filters are initially impeded from infiltrating deeper soil layers (Mohamed, 2023; Ajibade et al., 2024). However, once broken down into micro- or nanoparticles, CBMPs can migrate across soil matrices, especially via macropores, fractures, or root channels (Turner et al., 2024; Mohamed, 2023). The hydrophobicity and surface charge of CBMPs typically influence their interactions with mineral surfaces and colloids, which subsequently influence their environmental mobility or retention (Turner et al., 2024; Mohamed, 2023; Ajibade et al., 2024). CBMPs negatively impact soil texture, organic matter content, and pore water ionic strength (Mohamed, 2023; Ajibade et al., 2024).

Moreover, littering of CBs in the urban environment is a major environmental threat to the quality of urban water, owing to contamination from various hazardous chemicals contained in them, such as the rapid release of heavy metals and nicotine (Yang et al., 2024; Shen et al., 2021). Studies have demonstrated that during peak discharge measures, the flux of floating plastics, including CBMPs, can increase several-fold, highlighting the pivotal role of hydrology in plastic transport dynamics (Yang et al., 2024; Turner et al., 2024).

Urban and recreational areas, as well as sites near mining and agricultural operations, tend to exhibit higher concentrations of MPs and associated contaminants (Yogaswara et al., 2024; Díaz-Mendoza et al., 2024; Turner et al., 2024). Cumulative upstream contributions and lower dilution capacity often raise contaminant levels in downstream rivers and lakes (Novotny et al., 2009; Shen et al., 2021). Geographic variation emphasizes the need for a hydrological framework when assessing MP fate and mobility from CBs (Novotny et al., 2009; Michielssen et al., 2016). Once CBMPs find their way into aquatic bodies, owing to their buoyancy and tiny size, they cluster in sediments or remain in the water column, depending on hydrodynamic conditions (Wright et al., 2015; Belzagui et al., 2021). Estuaries and deltas are particularly vulnerable, as tidal pumping and flow variations prolong MP residence time and increase exposure risks (Freire Lima et al., 2021). CBs can also serve as sources and sinks for other environmental contaminants due to their high sorption capacity (El Hadri et al., 2021). According to research, 43.9–93% of fish and marine zooplankton contain microplastic fibers from CBs, which allow for bioaccumulation through trophic transmission (Shen et al., 2021). This leads to disruptions in osmoregulation, reproduction, and predator avoidance behaviors among aquatic organisms, threatening ecosystem stability (Yadav et al., 2024; Biefel et al., 2024). Given the prevalence of CB pollution and its transformation into MPs, hydrological risk evaluations and pollution mitigation measures must be re-examined. Understanding the transport mechanisms and the environmental fate of CB-MPs is crucial for developing effective solutions to safeguard water quality, aquatic biodiversity, and public health.

4.4. The fate of CBMPs in aquatic ecosystems

Complex systems comprising transport, deposition, transformation, and degradation define the fate of CBMPs in aquatic environments (Yang et al., 2024; Hon, 1977). Cigarette filters, primarily made of CA (Joly et al., 2018; Hon, 1977), are a major source of MPs entering aquatic systems through direct littering, wind movement, and urban runoff due to improper disposal. Upon entering aquatic habitats, CBMPs are subjected to hydrodynamic pressures, causing them to disperse throughout sediments, the water column, and surface waters (Freire Lima et al., 2021; Sharma et al., 2025). In these environments, CBMPs degrade into

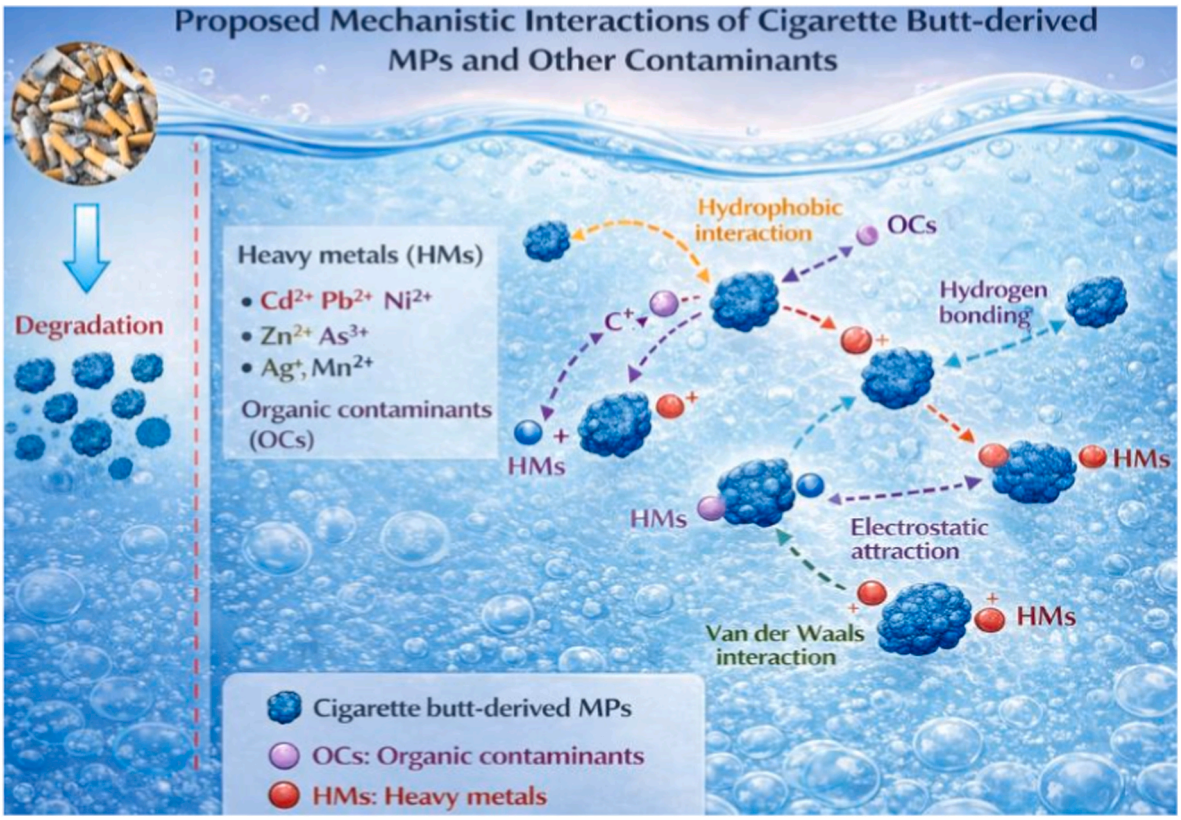


Fig. 5. Proposed mechanistic interactions of cigarette butt-derived MPs and other contaminants.

smaller particles through processes such as photodegradation, mechanical fragmentation, and limited biodegradation (Hon, 1977; Grīnbergs et al., 2021). Wave action and sediment abrasion contribute to physical breakdown, while UV radiation induces polymer fragmentation (Hon, 1977). Although microorganisms can partially degrade cellulose acetate, the process is often slow in aquatic environments (Green et al., 2020; Ach, 1993). This partial biodegradation releases microplastic fibers along with harmful substances like nicotine, ethylphenol, and heavy metals (Venugopal et al., 2021; Rebischung et al., 2018a).

Aquatic species are at risk from this contamination since they may ingest CBMPs, mistaking them for food. This can lead to physical obstructions, reduced growth, and increased mortality (Parker et al., 2017; Green et al., 2022). Since MPs can bioaccumulate in aquatic organisms and transfer through trophic levels, they pose a risk of food chain contamination and potential human exposure through seafood consumption (Ji et al., 2020). Beyond consumption, CBMPs also serve as carriers for microbes and organic contaminants, thus aggravating environmental degradation (Sharma et al., 2025). Even though the full extent of this combined toxicity is unknown, the combination of hazardous chemicals from tobacco stubs and microplastic fibers appears to have increased mortality in aquatic life (Green et al., 2020; Shen et al., 2021). Fig. 4 illustrates the fates and environmental hazards of cigarette butts.

One cigarette butt is said to spew hundreds of microplastic fibers every day, which could remain in the environment for decades (Belzagui et al., 2021; Shen et al., 2021). Understanding their distribution is crucial for assessing long-term ecological impacts and developing sustainable mitigation strategies. As a result, the European Union recently designated cigarette filters as single-use plastics due to their toxic features and widespread environmental harm (see Table 2) (Everaert et al., 2023).

4.5. CBMPs as vectors for contaminant exposure

MPs generated from CBs have demonstrated a strong capacity to act as vectors for various environmental contaminants (Turner et al., 2024; Rochman et al., 2013). MFs generated from the smoked filters (SFs) of CBs, for instance, might serve as vectors for a range of hazardous substances (Desideri et al., 2019). Cigarette filters act as partial barriers that trap chemicals during smoking; however, once discarded, they release these substances into the environment (Green et al., 2020). Whereas some chemicals leach slowly over time, others are rapidly released, posing immediate threats to both soil and aquatic ecosystems (Green et al., 2020; Mušič et al., 2023). Separate studies have found that the leachate produced by burning cigarette filters has high Chemical Oxygen Demand (COD) values, ranging from 2100 to 11,300 mg/L (Hossaini Motlagh et al., 2024).

Studies show that cigarettes leave behind almost 7000 chemicals, including heavy metals, nicotine, and PAHs, a large proportion of which have mutagenic or carcinogenic effects (Zeng et al., 2023). One of the major concerns is the adsorption of persistent organic contaminants (POPs) onto MPs produced from cigarettes. These MPs can act as concentrated reservoirs of harmful chemicals, often accumulating them in quantities much higher than those in the ambient water. For instance, a class of flame retardants called polybrominated diphenyl ethers (PBDEs) has been found to gather on MPs and then be passed on to aquatic life upon consumption (Sun et al., 2019). For example, research on Japanese medaka (*Oryzias latipes*) found that fish exposed to marine plastic particles had significantly higher PBDE levels than the control groups (Rochman et al., 2013).

This suggests that CBMPs can also play a vector role for PBDEs or similar contaminants in aquatic organisms. Furthermore, CBMPs have been found to adsorb heavy metals like iron and zinc, as revealed through atomic absorption spectroscopy analysis of CB leachates (Díaz-Mendoza et al., 2024). These metals adsorb onto MPs, therefore

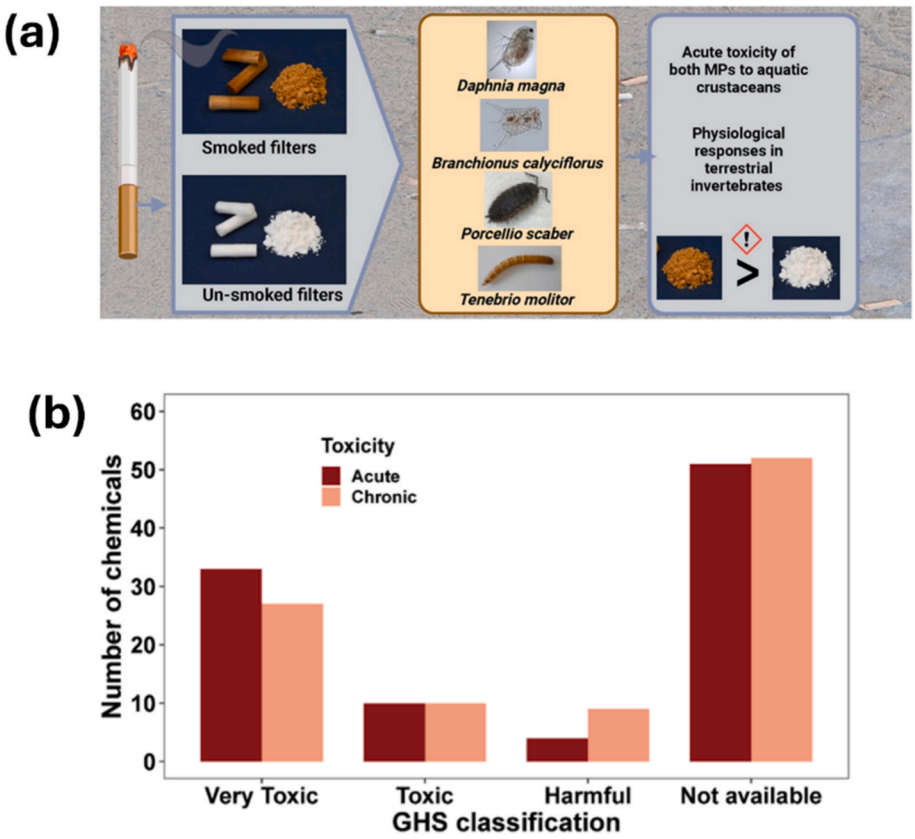


Fig. 6a. (A). Effects of Smoked and Unsmoked Cigarette Filters on the aquatic species of Model Organisms (Dolar et al., 2025) with permission. Copyright 2025 Elsevier. b. Globally Harmonised System of Classification and Labelling of Chemicals (GHS) Categorization for Aquatic Toxicity of Cigarette Butt Leachate Chemicals (Venugopal et al., 2021) with permission. Copyright 2021 Tobacco Regulatory Science

enhancing their bioavailability to marine life and raising concerns about the bioaccumulation and biomagnification through food webs (Venugopal et al., 2021; Hon, 1977). CBMPs function similarly to conventional MPs in their interactions with contaminants, often binding contaminants through hydrophobic forces, electrostatic interactions, and π-π bonding. These interactions are regulated by surface weathering, particle size, pH, and dissolved organic matter (Belzagui et al., 2021; Turner et al., 2024).

The fragmented structure and chemical makeup of CBMPs further enhance their contaminant binding capacities, making them hazardous when ingested by aquatic organisms (Dolar et al., 2025; Turner et al., 2024). Fig. 5 illustrates the mechanisms by which CBMPs interact with environmental contaminants, highlighting their significant role in contaminant mobility and toxicity. Weathering has been identified as one of the major factors that can potentially influence the characteristic features of plastic particles, such as their surface charge and structural modification (Adeleye, 2024). During weathering processes, variations in charge likely influence and facilitate the sorption affinity of plastic particles with certain contaminants (such as heavy metals, POPs, etc.), thereby aggravating their vector roles in the environment (Mušič et al., 2023; Adeleye, 2024). Recent studies have shown that MPs interactively bind with hydrophobic contaminants from contaminated soil and water (Turner et al., 2024; Yang et al., 2023a). Therefore, this interaction is another reason why plastic particles generated from any plastic-related products may be concerning and heighten environmental threats, with adverse effects on human health and various species (Sharma et al., 2025; Turner et al., 2024).

4.6. CB-derived leachate: toxicity and implications for aquatic organisms

CB leachates have been associated with significant metal pollution and phytotoxicity in both coastal and aquatic environments (Venugopal et al., 2021; Garrido Lazo et al., 2024). Given that CBs are the most commonly found garbage on beaches worldwide, their inappropriate disposal introduces various toxic chemicals into aquatic ecosystems (Shen et al., 2021). Leachates from CBs pose both acute and chronic risks to aquatic life (Sharma et al., 2025). The toxicity of the CB is determined by its form; the most hazardous are smoked cigarettes (SCB) containing tobacco residue, followed by smoked filters (SF) devoid of tobacco and unsmoked filters (USF). Lethal concentration (LC50) for freshwater and marine fish species varies, with SCBs proving harmful at concentrations as low as one butt per litre (Slaughter et al., 2011).

Numerous aquatic organisms are impacted by the effluent from CBs, including fish, algae, and invertebrates, particularly water fleas, which include *Daphnia magna* and *Ceriodaphnia* cf *dubia*, and marine bacteria like *Vibrio fischeri*. (Slaughter et al., 2011; Venugopal et al., 2021). In stagnant water bodies, like ponds and slowly flowing streams, CBs can release dangerous substances over extended periods, significantly impacting the aquatic ecosystem (Slaughter et al., 2011; Sharma et al., 2025). To safeguard aquatic environments, it is imperative to develop efficient waste management to address CB pollution. Recent discoveries of microplastic fibers emitted by CBs have also revealed increased ecotoxicity to aquatic species, such as fish, mollusks, and copepods (Yang et al., 2023a; Garrido Lazo et al., 2024). Additionally, research has demonstrated that leachates from CBs have detrimental impacts on fish populations, highlighting the need for comprehensive environmental risk assessments (Sharma et al., 2025; Garrido Lazo et al., 2024).

4.7. Ecotoxicological effects of CBs-derived leachate

CBs comprise a broad spectrum of chemicals derived from both tobacco and those generated during inhalation (Sharma et al., 2025).

Table 5
Evidence-based toxicity of cigarette butt leachates to aquatic organisms, including species, cigarette butt type, exposure concentration, duration, and ecotoxicological endpoints.

Study/Species	CB Type	Exposure Concentration	Exposure Duration	Endpoint	Effect/Metric	Reference
Freshwater fish	Smoked CB	1 CB/L	48 h	Mortality	LC50 ≈ 1 CB/L	Slaughter et al. (2011)
Daphnia magna	Smoked filter (SF)	0.017–0.620 SF/L	48 h	Immobilization	EC50 = 0.017 SF/L; EC100 = 0.620 SF/L	Belzagui et al. (2021)
Xenopus laevis embryos	Regular CB (RCB)	<1 CB/L	96 h	Developmental toxicity	NOAEC <1 CB/L; LOAEC <1 CB/L	Parker et al. (2017)
Xenopus laevis embryos	Menthol CB (MCB)	<1 CB/L	96 h	Developmental toxicity	Teratogenic effects	Parker et al. (2017)
Xenopus laevis embryos	E-cigarette CB (ECB)	8 CB/L	96 h	Developmental toxicity	NOAEC >8 CB/L; LOAEC >8 CB/L	Parker et al. (2017)
Intertidal mollusk Austrocochlea porcata	Smoked CB	5 CB/L	8 days	Mortality	100% mortality	Booth et al. (2015)
Nerita atramentosa	Smoked CB	5 CB/L	8 days	Behaviour	Increased activity at high CB concentrations	Booth et al. (2015)
Copepods (coastal species)	Smoked CB (leachate)	0.1 CB/L (water); 0.01 CB/L (sediment)	Acute–chronic	Survival/ reproduction	Significant adverse effects at environmentally realistic levels	Freire Lima et al. (2021)
Marine bacteria (Vibrio fischeri)	Smoked CB (leachate)	<1 CB/L	15–30 min	Bioluminescence inhibition	Strong acute toxicity response	Venugopal et al. (2021)
Rotifer (Brachionus calyciflorus)	SCF-MPs	10 mg L^{-1}	48 h	Survival	Significant mortality; higher toxicity than UCF-MPs	Dolar et al. (2025)
Freshwater cladocerans (mixed species)	CB leachate	0.2 CB/L	Chronic	Behaviour/growth	Behavioural impairment and developmental delays	Venugopal et al. (2021)

These substances can seep from discarded CBs, thereby endangering different ecosystems and potentially bioaccumulating in human food supplies as well as animals (Booth et al., 2015).

Leaching from filters and residual tobacco can result in toxic compounds contaminating surface waters and possibly infiltrating groundwater through poorly maintained landfills (Slaughter et al., 2011). This emphasizes the wide-ranging effects of improperly disposed cigarette waste, which goes far beyond mere littering concerns. Among 98 identified compounds in CB leachate, one-third are categorised as very harmful to aquatic life, and 10% are considered acutely or chronically toxic (Venugopal et al., 2021). The leachate contains a complex mixture of chemicals, with major harmful components including metals, phthalates, nicotine, PAHs, and volatile chemical compounds (Venugopal et al., 2021; Garrido Lazo et al., 2024). Even at low concentrations consistent with real-world environmental conditions (0.2 butts/L), CB effluent has been shown to induce carcinogenic effects, developmental delays, and behavioural changes in aquatic organisms (Dobaradaran et al., 2020).

Field research on the beaches of Santos identified CBs from 22 different brands in various stages of decomposition during both summer and winter seasons. The settling studies revealed the CB distribution in both water and sediment (Freire Lima et al., 2021). Furthermore, using *Xenopus laevis* embryos as a model, a study assessed the developmental toxicity of leachates from regular cigarette butts (RCB), menthol cigarette butts (MCB), and electronic cigarette butts (ECB) (Parker et al., 2017). In another study, exposure to CB leachate at a concentration of 5 butts per litre for 8 days resulted in 100% mortality among common intertidal mollusks. *Austrocochlea porcata* showed the highest sensitivity, exhibiting significant mortality at even lower concentrations (10% and 25%), potentially affecting its population dynamics. (Booth et al., 2015). Moreover, both smoked and unsmoked cigarette filters have been demonstrated to harm a variety of aquatic and terrestrial species, including *Daphnia magna, Brachionus calyciflorus, Porcellio scaber, and Tenebrio molitor*, highlighting impairment to physiological health with adverse effects on the ecological sphere (Fig. 6a) (Dolar et al., 2025).

Furthermore, a recent research output investigated the broad ecotoxicity of CB leachate using aquatic organisms. The authors analytically identified the highly toxic chemicals present in the leachate with their associated adverse effects (Venugopal et al., 2021). Based on the Globally Harmonised System (GHS) classification and the available Hazardous Concentration (HC) values reported in the literature (Fig. 6b). Altogether about 98 chemicals were systematically analyzed based on GHS classification and HC values from literature (Fig. 6b). Overall, leachates from CBs pose a major and usually underestimated environmental hazard. The wide range of effects emphasizes the urgent need for targeted mitigation strategies to address this pervasive contaminant that continuously threatens the delicate balance of aquatic and terrestrial ecosystems. Table 5 synthesises experimentally derived toxicity thresholds for cigarette butt leachates across freshwater and marine organisms, highlighting species-specific sensitivity, exposure duration, and ecotoxicological endpoints reported under environmentally relevant concentrations.

5. Analysis and characterisation of CBMPs

The characterisation of MPs generated from CBs requires a versatile approach including several advanced analytical techniques, as detailed below.

5.1. SEM

Scanning electron microscopy (SEM) has been the main tool used to study the morphological features of MPs obtained from CBs. For example, in a recent study examining the inflammatory effects of MPs from CBs on peripheral blood mononucleated cells, 2000x-resolution images analyzed with ImageJ were used to determine average fibre diameters. This was achieved utilizing measurements made at several points along several fibres visible in the SEM pictures. Furthermore, studies have shown variations in fibre diameters both inside individual filters and between brands. High-resolution SEM imaging has also revealed the complex architecture of CA fibres found in cigarette filters (Soltani et al., 2025).

Using SEM, the surface properties and internal structure of untreated (control) CA, smoked cigarette filters, and intermediate-stage biodegradable filters were investigated in another study. By means of a concentrated electron beam, this sophisticated microscopy approach generates high-resolution images of the surface of the filter, therefore facilitating the assessment and comparison of structural changes occurring during the degradation process (Parida et al., 2025). Each CB reportedly releases about 100 small microfibers (measured less than 0.2 mm) daily, as earlier published (Belzagui et al., 2021). Highly detailed photographs of surface features revealed by the SEM confirmed that MPs

were generated from CBs.

5.2. IR (FTIR and TG-FTIR)

The pyrolysis behaviour of mixed MPs, particularly those generated from CBs, has been investigated using thermogravimetric analysis in concert with Fourier-transform infrared spectroscopy (TG-FTIR). This approach offers important new perspectives on the thermal degradation behaviour of these materials and the emission of volatile chemicals during degradation. Thus, using TG-FTIR, the kinetics of pyrolysis have been investigated together with the chemical components liberated during thermal degradation (Yousef et al., 2024). Usually, the acquired crucial data tends to assist in developing suitable mitigation approaches, waste management, and recycling plans for these microplastic fibers from CBs. Moreover, by tracking the development of several functional groups using FTIR spectra at several phases of the pyrolysis process, researchers can now gain new insights of the properties of the produced volatile molecules and the fundamental processes of degradation mechanisms. Especially in relation to cellulose-based MPs, TG-FTIR can thus clarify the thermal stability of cellulose acetate and the generation of possibly harmful compounds during its decomposition (Yousef et al., 2024; Grīnbergs et al., 2021). Moreover, the technique can help to detect the microfibers generated by cigarette remnants, a major source. Attenuated Total Reflection Fourier-Transform Infrared (ATR-FTIR) spectroscopy has proven its usefulness as a robust approach for the characterisation of the chemical composition of CBMPs (Yogaswara et al., 2024).

ATR-FTIR spectrometers are known for their ease of use and minimal sample preparation, making them suitable for routine investigation of environmental samples (Belzagui et al., 2021; De Cesaris et al., 2024). Characteristic bands linked with the acetyl group at wavenumbers of 1240, 1370, and 1750 cm^{-1} have been shown to reflect the cellulose acetate composition present in tobacco filters (Yogaswara et al., 2024). This method has also been applied to examine the breakdown of CA in the discarded CBs, therefore offering important details on the environmental permanence of these materials. The non-destructive feature of ATR-FTIR makes it particularly helpful for tracking changes in the chemical structure of CB MPs over time (Araújo et al., 2022). Furthermore, ATR-FTIR can identify MPs smaller than 20 μm, and has demonstrated sensitivity to particles as small as 6 μm, making it especially useful for studying minute MPs from cigarette filters (Othman et al., 2023).

5.3. EDS

In a recent study, energy-dispersive X-ray spectroscopy (EDS) was used for elemental analysis, confirming the presence of heavy metals on sample surfaces. For more precise quantification, Inductively Coupled Plasma Mass Spectrometry (ICP-MS) was also employed, revealing a total heavy metal concentration of 629.7 μg/L across 14 elements. Leachate analysis from cigarette butts have uncovered several poisonous compounds, including anthracene (0.40 ng/L) and naphthalene (92.12 ng/L) (Yang et al., 2024).

5.4. HPLC-MS/MS

High-Performance Liquid Chromatography combined with Mass Spectrometry (HPLC-MS/MS) has been used to detect PAHs in cigarette butt leachates. These results highlight the possible environmental risks posed by harmful chemicals leaching from MPs generated by CBs (Yang et al., 2023a).

5.5. Zeta potential

Zeta potential measurements have been employed to understand the surface charge characteristics of cigarette butt-derived MPs. A study investigating the fate of nanoplastics in the environment reported that polystyrene-based nanoplastics with similar zeta potentials (−45 to −40 mV), but different sizes (50-800 nm) and morphologies, showed considerable sorption capacities when interacting with cigarette filters (El Hadri et al., 2021). This finding highlights the potential role of CBs in the transport and accumulation of other types of microplastics in the environment.

5.6. XPS and Raman spectroscopy

Emerging techniques such as X-ray photoelectron spectroscopy (XPS) and Raman spectroscopy can provide detailed information on the molecular architecture and surface chemistry of CBMPs (Othman et al., 2023; Dong et al., 2020). These methods could provide supplementary information to FTIR and EDS, allowing for a more complete characterisation of the complex nature of CB-derived contaminants.

5.7. GPC

Gel Permeation Chromatography (GPC) has also been applied to detect cellulose acetate remaining in composted CB samples, revealing insights into the biodegradation process of cigarette filters (Parida et al., 2022; Alkhoury et al., 2024). GPC analysis has confirmed the absolute lack of CA peaks in the composted samples, indicating the biodegradation under controlled conditions (Parida et al., 2022). This useful knowledge is crucial for devising effective solutions to ameliorate the environmental impacts of MPs generated from CBs and for influencing legislative decisions designed to minimise cigarette butt pollution.

6. Economic implications

The economic implications of cigarette plastic waste on a global scale are substantial, amounting to approximately US$26 billion, which includes US$20.7 billion attributed to damage to marine ecosystems and US$5 billion allocated for waste management (Sy, 2024). The financial burden of addressing CB-related litter falls heavily on taxpayers and has broad implications for national economies. For instance, in China, the annual cost is approximately US$2.6 billion; in India, it exceeds US$760 million; in Brazil, it surpasses US$200 million; in Germany, it is greater than US$230 million; and in Australia, it amounts to A$73 million (Green et al., 2023; CU, 2019). Moreover, research conducted in 2020 revealed the financial cost implications of cleaning up cigarette butt in the thirty largest US cities. The authors reported annual expenses ranging from US$4.7 million to US$90 million per city, with an average per capita cost estimated at US$6.46. Collectively, these thirty cities spend nearly US$264.5 million annually on CB clean-up (Schneider et al., 2020).

CB pollution imposes economic costs beyond the United States. In the United Kingdom, projected clean-up expenses reached £40 million (US $55 million)(UK, 2021). France allocates about €100 million (US$115 million) annually to manage CB-related waste (MacKenzie et al., 2021). These numbers underline the significant financial burden accrued on different countries. Beyond only direct clean-up and disposal expenses, cigarette butt pollution has additional economic consequences. For example, in Australia, discarded cigarettes were estimated to have caused 4558 fires in 2014-2015, resulting in damages of $80.8 million. Excluding bushfire-related costs, a 15-year Australian study (2003–2017), found that smoking materials, including cigarettes and pipes, accounted for 18% (161out of 900) of home fire-related deaths (Coates et al., 2019). Similarly, the 2012 Alt Empordà wildfires in Catalonia, Spain, highlight the severe consequences of improperly discarded CBs. With almost 13,800 ha of land burnt, this terrible event caused significant environmental and financial damage as well as four deaths. Investigations found that the main source of these fires was improperly discarded CBs (Anderson, 2024).

In the United States, negligent disposal of CBs and e-cigarette lighters

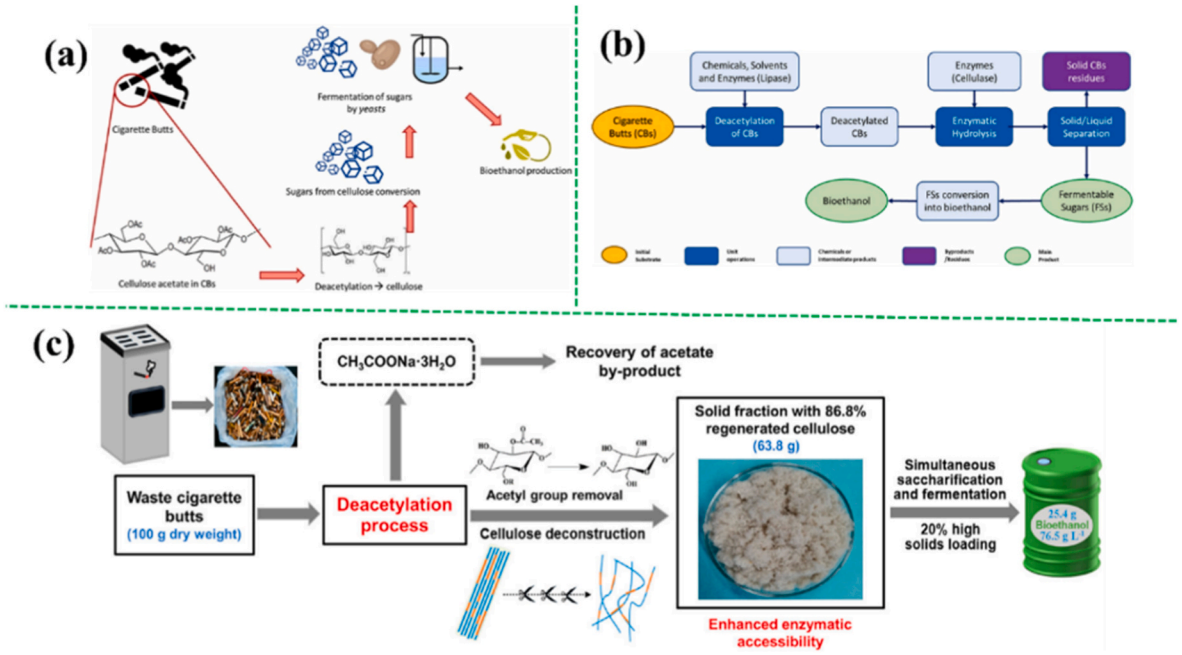


Fig. 7. (A) A circular biorefinery strategy for the valorisation of cigarette butts into sugars and bioethanol. (b) Operational unit for the valorisation of cigarette butts (Battista et al., 2023) with permission. Copyright 2023 Elsevier. (c) Production of Bioethanol using waste cigarette butts (Shen et al., 2020) with permission. Copyright 2020 ACS.

is a major fire risk, contributing to an estimated 130,000 wildfire events. These fires led to firefighting expenses of around US$2 billion and property losses of almost US$6 billion, reflecting serious financial repercussions (Schneider et al., 2020). To address this, some European countries, such as Denmark, embraced Extended Producer Responsibility (EPR) to enforce the polluter-pays principle (Diggle et al., 2020). However, beyond the EPR approach, more environmentally friendly alternatives are needed to manage these emerging plastic-related contaminants. For instance, a report commissioned by the World Wide Fund for Nature-Australia suggests that a levy of $0.004 per cigarette could raise $71 million per year to fund a product stewardship scheme aimed at reducing cigarette butt litter (Cleanup.org.au). Such initiatives represent a step towards internalizing the external costs of tobacco product waste and shifting the burden of responsibility onto the industry that profits from these products. However, an innovative approach toward the circular economy and materials revolution offers an eco-friendly pathway to address the menace of CBs.

We thus suggest biorefinery as a realistic and sustainable solution for managing CB waste. This approach not only helps address waste management issues but also enables the extraction of valuable components from CBs, converting them into useful materials. Further details of such innovative ideas are discussed in the next sections below.

7. Sustainable solutions to manage cigarette butts

Recent climate change, partially attributable to plastic waste, coupled with growing public awareness, necessitates a shift from a linear plastic economy to a more sustainable and circular model of management (Oladipupo et al., 2024; Choudhary et al., 2024). In this paradigm, the recycling and upcycling of plastic waste are imperative. Consequently, determining effective strategies to accomplish this objective represents one of the principal challenges (Oladipupo et al., 2024). CB wastes remain the most ubiquitous and recognized form of plastic pollution globally, posing considerable environmental and health hazards (Laurenza et al., 2021). Therefore, as awareness of their impact increases, novel solutions are required to tackle this widespread issue. Sustainable strategies need to be designed to alleviate environmental degradation and avert microplastic generation and contamination from CBs. Given this, we hereby present and outline significant innovative and sustainable solutions below.

7.1. Cellulose acetate recovery and composite sorbents

As knowledge of the environmental effects of these ubiquitous contaminants rises, mitigating techniques for controlling CBMPs become ever more crucial. To prevent generation of MPs and add value to waste, a recent study developed a new, green, and effective extraction technique to produce CA fibres from CBs. The approach involved washing CBs in hot water at 50 °C for 60 min, followed by multiple rinses in cold water and ethanol. Afterward, the resultant CA fibres were then dried at 60 °C for 1 h (De Fenzo et al., 2020). This approach not only mitigates contaminants but also maintains the purity of the obtained CA. The authors affirmed that the fabricated CA looks similar to pure CA fibres used in cigarette filters and could address CB degradation and leakage into the environment. Another study innovatively employed the recovered CA as a precursor platform for synthesizing of a versatile adsorbent subsequently used in water treatment. After using hot water and ethanol for cleaning the filters, porous cylinder-shaped crystals were produced. With removal yields above 79.6%, these cryogels augmented with activated carbon, were applied remove organic contaminants from water (De Cesaris et al., 2024).

This approach not only reduces the environmental impact of CBs but also offers a means to address water contamination. Several analytical approaches, including UV and ATR-FTIR spectroscopy, TGA, and DSC, confirmed the efficiency of this process (De Cesaris et al., 2024). Using a phase inversion technique, another study sustainably recycled CBs and filters into CA membranes. With removal rates of 85–90%, the resultant membranes demonstrated great efficiency in extracting heavy metals (chromium, cadmium, and lead) from aqueous solutions. This method not only produces valuable materials for water treatment but also presents a viable solution for CB pollution (Torkashvand et al., 2022).

7.2. Biorefinery: a potential innovative solution to CBs

Although much research has focused on policy and public sensitisation regarding improper disposal and its environmental consequences, implementing innovative solutions is necessary. As nations grapple with the pervasive impact of CBs, new approaches offer sustainable, environmentally benign solutions to turn this waste stream into value-added resources. Therefore, this section focuses on innovative approaches to transform CBs into valuable materials. Biorefinery offers a novel method to solve the persistent problem of cigarette butt pollution, generate value-added materials, and support the transition toward a circular economy and materials revolution, all while prioritizing environmental cleanliness. For instance, a recent study has investigated the potential of biorefinery systems to generate fermentable sugars and bioethanol, providing a sustainable pathway to turn hazardous waste into value-added material towards the materials revolution (Battista et al., 2023).

Another study subjected the spent CBs to various pretreatment approaches to obtain CA fibre. The obtained fibres were then enzymatically treated to produce fermentable sugars. The authors further indicated that adding cellulase at 10% of the CBs concentration yielded promising results. During microbial fermentation, these fermentable sugars served as a viable substrate for bioethanol production. This eco-friendly approach not only produces bioethanol, useful in biofuel and chemical industries, but also helps mitigate CB pollution through the biorefinery process (see Fig. 7) (Battista et al., 2023).

A recent study also used CBs as a cellulosic precursor in a biorefinery process employing a hydrothermal strategy with phosphoric acid as a catalyst to produce levulinic acid, a key precursor in the production of biofuel (Laurenza et al., 2021). The approach has also been extended to generate triacetin from CBs via pyrolysis. This is another way of addressing limited resources, as wastes are being converted into value-added materials (Yousef et al., 2023). Furthermore, with the aid of pyrolysis or gasification, syngas, bio-oil, and biochar have also been reportedly generated from CBs (Wang et al., 2020). Among the benefits of using biorefinery systems are their scalability and integration into existing waste management frameworks. Establishing specialised facilities for processing CBs tends to help in building a sustainable loop capable of turning these hazardous wastes into useful resources. In valorising waste, this reduces environmental load and landfill pressure while opening financial opportunities (Adeleye et al., 2023).

As studies and pilot projects continue to demonstrate feasibility, biorefinery presents a sustainable and practical solution to global cigarette butt pollution. (Battista et al., 2023). Beyond diverting CBs from waterways and landfills, their collection and processing could generate financial incentives. Developing scalable, cost-effective technologies for CB valorising is crucial to advancing the circular economy and materials recovery efforts.

7.3. Integrated solutions for the environmental impacts of CBMPs

While policy interventions and public awareness are essential in dealing with the CBMPs, developing integrated techniques with sustainable features is equally important. Both individual and combined remediation techniques are necessary. Various techniques have been successfully employed for the removal of MPs in aquatic environments. Therefore, various removal methods, such as the membrane bioreactor (MBR) (Michielssen et al., 2016), adsorption (Adeleye et al., 2025; Wu et al., 2023), advanced oxidation processes (John et al., 2023), coagulation (Park et al., 2021), photocatalysis (John et al., 2023), and filtration (Akarsu et al., 2021) have been explored to mitigate MPs pollution. Among these, adsorption is considered a cost-effective method (Adeleye et al., 2025). For example, one study achieved 99.96% removal efficiency using magnetically modified biochar from rice husks, attributed to surface complexation and electrostatic interaction (Wu et al., 2023). Another study demonstrated that the membrane bioreactor (MBR) systems could remove over 99% of MPs from wastewater (Michielssen et al., 2016). However, it is important to state that the MBR approach is likely to experience membrane fouling, which can subsequently make the system expensive (Sol et al., 2020).

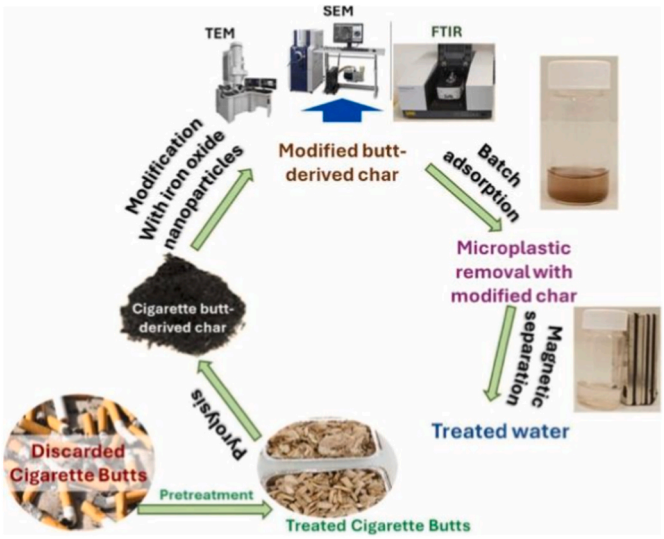


Fig. 8. Scheme for the synthesis pathway of nanocomposites from CBs and application for the removal of polystyrene microplastics (Adeleye et al., 2025) with permission. Copyright 2025 Elsevier.

In a recent study, Grbic and colleagues employed magnetic extraction using iron nanoparticles coated with hexadecyltrimethoxysilane, enhancing MPs isolation due to the magnetic synergy (Grbic et al., 2019). In coagulation, which uses electrostatic interactions between MPs (e.g., PE and PS) and the metal ions (e.g., Fe or Al), the removal efficiency was only 40%. Interestingly, replacing the conventional coagulant with plant-derived tannic acid improved removal efficiency to >95% for MP-PS beads through filtration (Park et al., 2021).

Recently, we employed discarded cigarette butts (DCBs) as precursors for the synthesis of cigarette butt char (CBC), which was further modified to produce TiO_2-based (TCBC) and magnetic (MCBC) nanocomposites (Fig. 8). These nanocomposites were investigated as adsorbents for polystyrene (PS) microplastic removal from water. Among the nanocomposites, MCBC demonstrated the highest PS removal efficiency (~87.6 %) and reusability over five cycles. Therefore, the study offers a circular, low-cost solution for mitigating plastic pollution and cigarette waste (Adeleye et al., 2025).

7.4. Extended producer responsibility (EPR): key effective strategy for cigarette butt waste management

EPR is widely acknowledged as a mechanism to ensure industries bear responsibility for their environmental impacts. Fundamentally, EPR schemes place end-of-life product management obligations on producers, often internalizing environmental externalities into product costs (Croes et al., 2023). At its core, an EPR scheme assigns producers the responsibility for managing their products at the end of their life-cycle, in some instances, with the effect of internalizing negative environmental externalities into the product's price (Diggle et al., 2020; Novotny et al., 2014). Extended Producer Responsibility (EPR) represents one of the most effective policy instruments for addressing cigarette butt (CB) and tobacco-related waste pollution, making producers accountable for environmental impacts across the full product lifecycle, including end-of-life management (Croes et al., 2023; Novotny et al., 2014). In the US bill, Federal lawmakers in February 2020 introduced the Break Free From Plastic Pollution Act. It enshrines extended producer responsibility for plastic bottles, packaging and plastic cigarette filters (Lowenthal, 2020). Generating revenue from producers to cover the end-of-life costs of managing their products and packages holds

Table 6
Comparative evaluation of mitigation strategies for cigarette butt pollution: Technology readiness level (TRL), cost, scalability, environmental risks, and social acceptability.

Mitigation Option	TRL	Relative Cost[a]	Scalability	Emissions/ Secondary Pollution Risk	Social Acceptance	Overall Highlights	Ref
Extended Producer Responsibility (EPR)/ collection programs	Medium (TRL 6–7)	Moderate	High	Low–Medium	Medium–High	EPR shifts cost to industry and can reduce litter entry points; includes collection infrastructure.	(Croes et al., 2023; Novotny et al., 2014)
Bans or regulatory policies (e.g., single-use plastics designation)	High (TRL 8–9)	Low–Moderate	High	Very low	Variable	Regulatory bans (as in EU plastics directive) reduce CB waste at source but require strong enforcement.	(Venugopal et al., 2021; Novotny et al., 2014)
Recycling/Upcycling (CA recovery, composite sorbents)	Medium (TRL 5–7)	Moderate	Medium	Low	Medium	Recovery of cellulose acetate (CA) into membranes, adsorbents, or composites shows promise; economic viability has been demonstrated for some applications.	Torkashvand et al. (2022)
Biorefinery/pyrolysis to fuels/chemicals	Medium (TRL 4–6)	Moderate–High	Medium	Medium	Low–Medium	Pyrolysis can recover triacetin or biofuels, but high energy input and by products must be managed.	Yousef et al. (2023)
Advanced sorbents & nanocomposites (e.g., MCBC for microplastic/ treatment)	Low–Medium (TRL 3–6)	Low–Moderate	Medium	Low	Medium–High	Lab-scale adsorbents show high efficiency, but full-scale deployment will require demonstration in environmental conditions.	Adeleye et al. (2025)

[a] Relative Cost: estimates derived from literature values where available; costs can vary by region and implementation scale. Technology Readiness Level (TRL) scale: 1 (basic concepts) to 9 (fully commercialized).

great potential to support the improvement of waste management for most consumer products (Diggle et al., 2020). EPR schemes compel tobacco companies to fund waste collection, recycling infrastructure, and disposal costs while providing incentives for filter redesign and eco-innovation (Croes et al., 2023; Novotny et al., 2014). Global momentum is accelerating, with EPR mandates featured prominently in the EU Single-Use Plastics Directive (Article 8) and draft UN Plastics Treaty negotiations, targeting 50% litter reduction by 2030 (Sy et al., 2025).

8. Comparative evaluation of mitigation strategies of cigarette butts and key trade-offs

8.1. Comparative evaluation

To translate mitigation concepts into actionable frameworks, we developed a decision matrix that evaluates key cigarette butt (CB) mitigation and circular economy options across six criteria: Technology Readiness Level (TRL), cost, scalability, emissions, secondary pollution risk, and social acceptance (Table 6). These criteria reflect not only environmental performance but also practical deployment considerations necessary for high-value policy and industry decision-making. EPR and regulatory bans tend to have the highest potential to reduce environmental inputs and are relatively low-cost to implement once mandated, but require robust governance and industry cooperation (Diggle et al., 2020; Sy et al., 2025). Collection and recycling/upcycling offer circular solutions, but economic feasibility and market demand for recovered materials remain key constraints. For example, recovered cellulose acetate is useable in membranes and fashion products, though quality and processing cost must be balanced (Torkashvand et al., 2022). Biorefinery and pyrolysis can valorize CBs into fuels or chemicals, but often involve high energy consumption and may generate secondary emissions if not carefully designed and regulated (George et al., 2022; Shen et al., 2020). Advanced sorbent materials and composites offer novel pathways for microplastic removal, but current development is largely at pilot or lab scale and requires further work for field deployment (Adeleye et al., 2025). By evaluating these technologies through a consistent lens, we provide readers with a clearer understanding of feasibility, risks, and potential trade-offs associated with each mitigation pathway. This framework supports balanced decision-making and identifies areas where further research and investment are needed.

8.2. Key trade-offs

Collection programs and public litter management are well-established and socially acceptable, but their effectiveness depends heavily on community participation and sustained funding (Zafeiridou et al., 2018). EPR schemes offer a policy-driven approach to internalize environmental costs and incentivize product redesign, but they carry risks of tobacco industry greenwashing unless safeguards are implemented, as highlighted in the EU and UN treaty contexts (Green et al., 2023). Bans and tobacco control policies remain the most effective in reducing CB generation at source and provide co-benefits for public health (Sy, 2024; Green et al., 2023). Recycling, upcycling, and sorbent-based removal offer promising circular economy and remediation pathways, but they are currently limited by technological readiness, cost, and potential secondary pollution if microplastics are not fully contained (Adeleye et al., 2025; Torkashvand et al., 2022; Shen et al., 2020). Pyrolysis and biorefinery approaches can recover energy or chemical feedstocks but are capital-intensive and not yet scalable (Janković et al., 2023; Premchand et al., 2025). Overall, an integrated strategy combining upstream regulatory measures (bans, EPR) with targeted downstream interventions (collection, recycling, remediation) is likely the most effective pathway for reducing CB pollution while balancing feasibility, cost, and environmental outcomes.

9. Standardization agenda for cigarette butt and filter microplastic research

A major limitation in the existing literature is the lack of harmonised protocols, which hampers comparability across studies and undermines meta-analyses. To address this, we propose a minimum standardization framework covering sampling, preparation, exposure simulation, and reporting.

9.1. Recommended protocol elements and suggested reporting checklist

1. Sampling and Source Characterisation

- ✓ Clear distinction between smoked, unsmoked, and partially smoked cigarette filters.
- ✓ Documentation of cigarette brand, filter composition (cellulose acetate content, additives), and presence/absence of biodegradable claims.
- ✓ Field-collected vs laboratory-prepared CBs explicitly stated.

2. Aging and Weathering Simulation
   - ✓ Use of standardized artificial aging protocols (e.g., UV exposure duration, thermal cycling, wet–dry cycles).
   - ✓ Reporting of aging intensity and equivalence to estimated environmental residence time.
3. Leachate Preparation
   - ✓ Standardized solid-to-liquid ratios, extraction times, and water chemistry (e.g., freshwater, seawater, simulated rainwater).
   - ✓ Clear reporting of whether leachates include dissolved chemicals only or suspended microplastic fragments.
4. Microplastic Characterisation
   - ✓ Consistent size classification (e.g., >5 mm, 1–5 mm, 100 μm–1 mm, <100 μm).
   - ✓ Analytical methods (FTIR, Raman, SEM) explicitly stated, including detection limits.
   - ✓ Reporting in harmonised units (e.g., particles per butt, mass per butt, particles per litre).
5. Toxicity and Exposure Studies
   - ✓ Explicit separation of physical (particle-driven) and chemical (leachate-driven) effects.
   - ✓ Inclusion of environmentally relevant concentrations alongside worst-case scenarios.
6. Environmental and Human-Relevance Context
   - ✓ Justification of experimental concentrations relative to field-measured CB densities.
   - ✓ Clear linkage between laboratory findings and realistic exposure pathways.

10. Research roadmap: priority questions for future studies

Building on the identified gaps, the following prioritised research questions are proposed to guide future investigations and policy-relevant science.

i. **Degradation Dynamics:** *How do cellulose acetate cigarette filters fragment into micro- and nanoplastics under realistic environmental stressors over time* (Belzagui et al., 2021; Suárez-Rodríguez et al., 2017; Croes et al., 2023)? Understanding the long-term fragmentation of cellulose acetate cigarette filters under realistic environmental stressors remains limited. Future studies should quantify micro- and nanoplastic generation under combined UV radiation, mechanical abrasion, and wet–dry cycling across seasonal to multi-year timescales to better reflect environmental exposure conditions.

ii. **Chemical–Physical Interactions:** *How does environmental aging influence the co-release of microplastics and sorbed toxicants from cigarette filters* (Venugopal et al., 2021; Yang et al., 2024; Novotny et al., 2009)? Environmental aging may alter filter surface chemistry, enhancing both microplastic release and toxicant mobilisation. Critical knowledge gaps remain regarding the relative contributions of particulate-bound versus dissolved fractions of nicotine, PAHs, and metals to observed toxicity, particularly following oxidative or photochemical weathering.

iii. **Smoked vs Unsmoked Filters:** *To what extent do smoked and unsmoked cigarette filters differ in microplastic release, leachate composition, and toxicity* (Slaughter et al., 2011; Zeng et al., 2023; Soltani et al., 2025)? Smoked filters exhibit markedly different chemical loads and surface properties compared with unsmoked filters, yet their comparative microplastic release rates, leachate toxicity, and sorption behaviour remain poorly constrained. Systematic side-by-side studies are required to disentangle combustion-derived effects from polymer aging alone.

iv. **Environmental Fate and Transport:** *What are the dominant transport pathways and environmental sinks of cigarette butt–derived microplastics* (Booth et al., 2015; Oceancare; Watson et al., 2016)? The transport pathways and environmental sinks of cigarette butt-derived microplastics across soils, stormwater, rivers, and coastal systems are insufficiently resolved. Future work should integrate field observations with hydrodynamic and particle-tracking models to identify dominant accumulation zones and long-range dispersal potential.

v. **Ecotoxicological Thresholds:** *What are environmentally relevant exposure concentrations and toxicity thresholds for CB-derived microplastics and leachates* (Díaz-Mendoza et al., 2024; Green et al., 2014; Freire Lima et al., 2021)? Despite extensive toxicity evidence, environmentally relevant exposure concentrations and chronic effect thresholds for CB-derived microplastics and leachates remain unclear. Multi-species, multi-trophic studies employing sub-lethal endpoints are essential for improving ecological risk assessments and regulatory relevance.

vi. **Human Exposure Relevance:** *Do CB-derived microplastics represent a distinct human exposure pathway compared with other environmental microplastic sources* (Novotny et al., 2011; De Cesaris et al., 2024)? Whether CB-derived microplastics represent a distinct human exposure pathway compared with other microplastic sources remains largely unexplored. Future research should assess inhalation, ingestion, and dermal exposure potential, considering unique additive profiles, particle morphologies, and co-occurring toxicants associated with cigarette filters (Novotny et al., 2011; Adeleye et al., 2025).

vii. **Remediation Effectiveness:** *How effective are current mitigation and remediation strategies in preventing secondary microplastic pollution from cigarette butts* (Ajibade et al., 2023; Adeleye et al., 2025; Grīnbergs et al., 2021; Nnadozie–Miklós Et Al., 2023)? The real-world effectiveness of mitigation strategies such as collection programmes, sorbent-based capture, upcycling, and thermochemical treatments remains uncertain. Long-term field evaluations are needed to determine whether these interventions genuinely prevent secondary microplastic release or merely shift environmental burdens.

viii. **Policy and Design Interventions:** *Can cigarette filter redesign or elimination meaningfully reduce environmental harm without unintended trade-offs* (Venugopal et al., 2021; Novotny et al., 2009; Everaert et al., 2023)? Cigarette filter redesign or elimination has been proposed as a pollution-prevention strategy, yet potential ecological, behavioural, and regulatory trade-offs remain under-examined. Interdisciplinary research is required to assess whether such interventions reduce environmental harm without undermining public health objectives.

ix. **Life-Cycle and Trade-Off Analyses:** *What are the life-cycle environmental, economic, and social trade-offs of different CB mitigation strategies* (Sharma et al., 2025; Choudhary et al., 2024; Kumar et al., 2024)? Comparative life-cycle assessments of CB mitigation options are scarce. Future analyses should integrate environmental, economic, and social dimensions to identify unintended consequences, rebound effects, or inequitable cost distributions associated with large-scale implementation of remediation or valorisation strategies.

x. **Extended Producer Responsibility (EPR) and Governance:** *Under what governance conditions can EPR schemes for cigarette filters effectively reduce pollution* (Croes et al., 2023; Sharma et al., 2025; Rebischung et al., 2018a)? While EPR schemes have been proposed for cigarette filters, their effectiveness depends strongly on governance design. Research is needed to evaluate conditions under which EPR reduces pollution without enabling

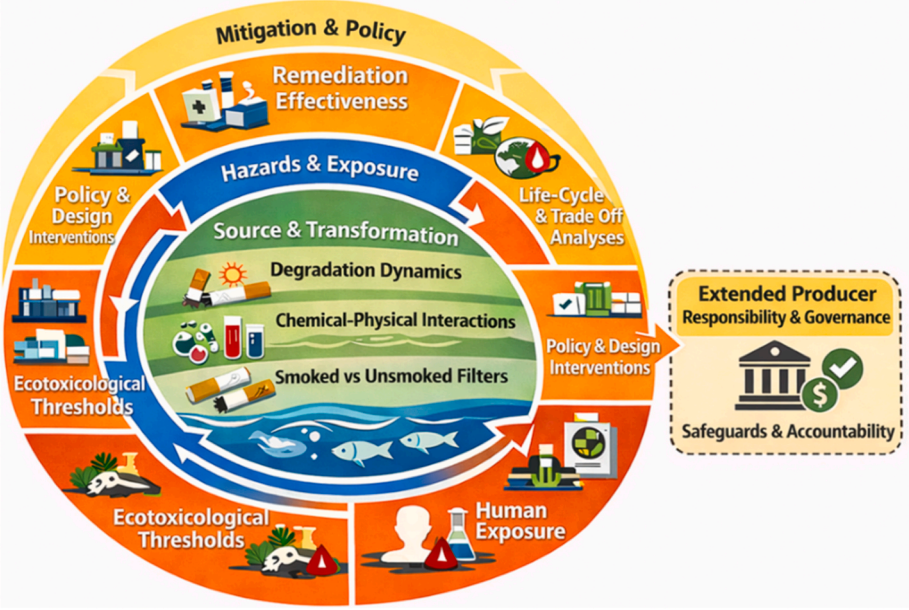


Fig. 9. Summarised research roadmap and priority questions for future studies.

greenwashing, regulatory capture, or conflicts with tobacco control frameworks such as the World Health Organization Framework Convention on Tobacco Control (WHO FCTC). The WHO FCTC is the first global public-health treaty, adopted in 2003, that aims to reduce tobacco use and exposure to tobacco smoke worldwide. It provides a legally binding framework for countries to implement tobacco-control measures (Craig et al., 2019). The above developed research roadmap and prioritised questions for future studies are summarised and presented in Fig. 9.

11. Conclusions

Cigarette butts are among the most widespread yet underregulated sources of plastic pollution worldwide. Composed predominantly of cellulose acetate fibers, they undergo fragmentation and weathering processes in the environment, generating cigarette butt-derived microplastics and nanoplastics that contribute significantly to secondary microplastic contamination across terrestrial, freshwater, and marine ecosystems. Transportation through urban runoff, stormwater pathways, and tidal exchange promotes their redistribution, with estuarine and coastal zones emerging as major accumulation areas. Beyond their long environmental persistence, cigarette butts represent chemically complex hazardous waste. Smoked filters leach a variety of toxic substances, including nicotine, polycyclic aromatic hydrocarbons, and heavy metals. At the same time, the associated microplastics act as multifunctional contaminant carriers capable of adsorbing and mobilizing additional contaminants in the environment.

Experimental evidence across diverse taxa has demonstrated acute and chronic toxicity at environmentally relevant concentrations, manifesting in developmental, behavioral, reproductive, and genotoxic effects. Advances in characterisation techniques have improved the understanding of their morphology, surface chemistry, interactions with contaminants, and degradation behavior. However, current methodological inconsistency limits comparability across studies and constrains the assessment of ecological risks. Development of standardized protocols for sampling, artificial weathering, and toxicity testing is necessary to improve reproducibility and support reliable environmental risk evaluation. Important knowledge gaps persist regarding long term degradation dynamics, contaminant release rates, transport modeling, human exposure pathways, and the establishment of environmentally realistic risk thresholds. The environmental and socio-economic consequences of cigarette butt pollution are extensive, encompassing ecosystem degradation, increased waste management demands, and infrastructure maintenance costs.

Addressing these impacts requires a strategic combination of upstream and downstream measures. Preventive approaches such as product design reform, regulatory controls, and extended producer responsibility offer the highest potential for source reduction. Complementary downstream strategies, including recycling, material valorisation, and site-specific remediation, can mitigate existing contamination and support circular economy principles. Robust governance frameworks are essential to ensure regulatory effectiveness and to prevent superficial or misleading sustainability claims. Overall, cigarette butt-derived microplastics represent a distinct priority category of microplastic pollution that integrates polymer persistence, chemical hazard, and socioeconomic burden. Coordinated scientific standardization, evidence-based policy development, and preventive management strategies are crucial to mitigate their long-term environmental and societal effects.

CRediT authorship contribution statement

Aderemi Timothy Adeleye: Writing – original draft, Visualization, Investigation, Data curation, Conceptualization. **Md Mezbaul Bahar:** Writing – review & editing, Visualization, Supervision. **Mallavarapu Megharaj:** Writing – review & editing, Supervision. **Cheng Fang:** Writing – review & editing, Supervision. **Mohammad Mahmudur Rahman:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No new data were generated or analyzed.

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